Exam

Name___________________________________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) Which is NOT considered to be a primary function of the respiratory system? 1) ________
A) protection against pathogens
B) vocalization
C) regulation of pH balance
D) regulation of water balance
E) gas exchange

2) Ventilation is also known as 2) ________
A) expiration.
B) breathing.
C) blowing.
D) air conduction.
E) inspiration.

3) The upper respiratory tract includes all EXCEPT which of the following? 3) ________
A) mouth B) nasal cavity C) lungs D) larynx

4) The lower respiratory tract includes 4) ________
A) all of the bronchial branches and the lungs.
B) all of the bronchial branches, the lungs, and the trachea.
C) all of the bronchial branches.
D) only the lungs.
E) only the trachea.

5) Pulmonary ventilation refers to the 5) ________
A) movement of dissolved gases from the blood to the interstitial space.
B) movement of dissolved gases from the interstitial space to the cells.
C) movement of dissolved gases from the alveoli to the blood.
D) movement of air into and out of the lungs.
E) utilization of oxygen.

6) Alveolar ventilation refers to the 6) ________
A) movement of air into and out of the alveoli.
B) movement of dissolved gases from the alveoli to the blood.
C) movement of air into and out of the lungs.
D) movement of dissolved gases from the blood to the alveoli.
E) utilization of oxygen by alveolar cells to support metabolism.

7) The sites of gas exchange within the lungs are 7) ________
A) bronchioles.
B) alveolar ducts.
C) alveoli.
D) terminal sacs.
E) pleural spaces.

8) Place the following structures of the respiratory tree in the order in which air passes through them. 8) ________
1. secondary bronchi
2. bronchioles
3. primary bronchi
4. alveoli
5. terminal bronchioles
A) 4, 1, 2, 3, 5 B) 3, 1, 2, 5, 4 C) 1, 3, 2, 5, 4 D) 1, 3, 5, 2, 4 E) 3, 1, 5, 2, 4

9) The airway between the larynx and the primary bronchi is the 9) ________
A) trachea.
B) laryngeal duct.
C) alveolar duct.
D) bronchiole.
E) pharynx.

10) The lungs are surrounded by ________ membranes. 10) ________
A) pericardial
B) pleural
C) pulmonary
D) costal
E) thoracic

11) The lungs are located in the ________ cavity. 11) ________
A) pleural
B) thoracic
C) costal
D) pulmonary
E) pericardial

12) Pressure and volume of gas in a container are related to temperature and number of gas molecules. This is known as ________ law. 12) ________
A) Ohm's
B) Dalton's
C) the ideal gas
D) Boyle's
E) Henry's

13) Type II alveolar cells 13) ________
A) allow rapid diffusion of gases through their thin membranes, secrete a chemical known as surfactant, and are phagocytic.
B) are phagocytic.
C) secrete a chemical known as surfactant.
D) allow rapid diffusion of gases through their thin membranes.
E) None of the statements are true.

14) Type I alveolar cells 14) _______
A) are phagocytic.
B) allow rapid diffusion of gases through their thin membranes.
C) allow rapid diffusion of gases through their thin membranes, secrete a chemical known as surfactant, and are phagocytic.
D) secrete a chemical known as surfactant.
E) None of the statements are true.

15) Which feature of the alveoli allows for the ease of diffusion of gases? 15) _______
A) Elastin fibers allow the alveoli to stretch thin enough for diffusion to occur.
B) Type II alveolar cells secrete surfactant.
C) They are made of a single layer of epithelium.
D) They are made of a single layer of simple squamous epithelium and elastin fibers allow the alveoli to stretch thin enough for diffusion to occur.

16) Surfactant 16) _______
A) is not found in healthy lung tissue.
B) protects the surface of the lungs.
C) phagocytizes small particulate matter.
D) helps prevent the alveoli from collapsing.
E) replaces mucus in the alveoli.

17) The common passageway shared by the respiratory and digestive systems is the 17) _______
A) pharynx.
B) esophagus.
C) vestibule.
D) glottis.
E) larynx.

18) When the diaphragm and external intercostal muscles contract, 18) _______
A) the volume of the thorax increases.
B) the volume of the thorax decreases.
C) expiration occurs.
D) the lungs collapse.
E) the volume of the lungs decreases.

19) Dalton's law states that 19) _______
A) the volume of gas that will dissolve in a solvent is proportional to the solubility of the gas and the gas pressure.
B) in a mixture of gases like air, the total pressure is the sum of the individual partial pressures of the gases in the mixture.
C) gas volume and temperature are directly proportional.
D) gas volume and pressure are inversely proportional.
E) None of the answers are correct.

20) Air moves into the lungs because 20) _______
A) the thorax is muscular.
B) contraction of the diaphragm decreases the volume of the pleural cavity.
C) the volume of the lungs decreases with inspiration.
D) the gas pressure in the lungs is less than outside pressure.
E) All of the answers are correct.

21) Air moves out of the lungs because 21) _______
A) the volume of the lungs decreases with expiration.
B) contraction of the diaphragm increases the volume of the pleural cavity.
C) the thorax is muscular.
D) the gas pressure in the lungs is less than outside pressure.
E) All of the answers are correct.

22) In quiet breathing, 22) _______
A) inspiration and expiration are both passive processes.
B) inspiration involves muscular contractions and expiration is passive.
C) inspiration and expiration involve muscular contractions.
D) inspiration is passive and expiration involves muscular contractions.
E) None of the answers are correct.

23) Boyle's law states that gas volume is 23) _______
A) inversely proportional to temperature.
B) directly proportional to temperature.
C) directly proportional to pressure.
D) inversely proportional to pressure.
E) None of the answers are correct.

24) Air entering the body is filtered, warmed, and humidified by the 24) _______
A) lungs.
B) lower respiratory tract.
C) alveoli.
D) upper respiratory tract.
E) parts of the upper and lower respiratory tract.

25) A typical value for intrapleural pressure is ________ mm Hg. 25) _______
A) 0 B) +3 C) +20 D) −20 E) −3

26) According to the law of LaPlace, when comparing two alveoli lined with fluid, pressure in the one with the ________ diameter will be greater. 26) _______
A) larger B) smaller

27) When alveolar pressure is equal to atmospheric pressure, air flows into the lungs. 27) _______
A) True B) False

28) Active expiration is produced by contraction of 28) _______
A) internal intercostals only.
B) diaphragm only.
C) abdominal muscles and internal intercostals.
D) abdominal muscles only.
E) external intercostals only.

29) When the diaphragm, external intercostal and scalene muscles contract, 29) _______
A) the volume of the lungs decreases.
B) expiration occurs.
C) intrapleural pressure decreases.
D) intrapulmonary pressure increases.
E) All of the answers are correct.

30) If a student inhales as deeply as possible and then blows the air out until he cannot exhale any more, the amount of air that he expelled is his 30) _______
A) minimal volume.
B) vital capacity.
C) expiratory reserve volume.
D) inspiratory reserve volume.
E) tidal volume.

31) Total cross-sectional area ________ with each division of the airways. 31) _______
A) decreases B) increases C) does not change

32) Blood vessels cover approximately ________% of the alveolar surface. 32) _______
A) < 10 B) 50 C) > 90 D) 80-90 E) 10-20

33) In the lungs, the 33) _______
A) blood flow rate is lower and the blood pressure is lower, than the blood flow rate and the blood pressure in other tissues.
B) blood flow rate and the blood pressure are the same as in other tissues.
C) blood flow rate is lower and the blood pressure is higher, than the blood flow rate and the blood pressure in other tissues.
D) blood flow rate is higher and the blood pressure is higher, than the blood flow rate and the blood pressure in other tissues.
E) blood flow rate is higher and the blood pressure is lower, than the blood flow rate and the blood pressure in other tissues.

34) The distance between the alveolar air space and capillary endothelium is ________, allowing gases to diffuse ________ between them. 34) _______
A) long, rapidly
B) short, slowly
C) long, slowly
D) short, rapidly
E) None of the answers are correct–distance does not affect diffusion rate.

35) Flow of air 35) _______
A) is directly proportional to a pressure gradient, and flow decreases as the resistance of the system increases.
B) is directly proportional to a pressure gradient, and flow increases as the resistance of the system increases.
C) is directly proportional to the resistance, and flow increases as the resistance of the system increases.
D) is directly proportional to the resistance, and flow decreases as the pressure of the system increases.
E) None of the answers are correct.

36) An increase in P_{CO_2} would cause the bronchioles to 36) _______
A) constrict and the systemic arterioles to dilate.
B) constrict and the systemic arterioles to constrict.
C) dilate and the systemic arterioles to dilate.
D) dilate and the systemic arterioles to constrict.
E) None of the answers are correct.

37) Chronic inhalation of fine particles that reach the alveoli leads to ________ lung disease. 37) _______
A) congestive
B) compliant
C) fibrotic
D) obstructive
E) restrictive

38) Poiseuille's law is summarized this way. 38) _______
A) $P_1V_1 = P_2V_2$ B) $PV = nRT$ C) $P = 2T/r$ D) $R \propto L\eta/r^4$

39) Histamine's primary role in the respiratory system is as a 39) _______
A) bronchoconstrictor.
B) surfactant.
C) vasoconstrictor.
D) vasodilator.
E) bronchodilator.

MATCHING. Choose the item in column 2 that best matches each item in column 1.

Match the lung volume with its description.

40) the additional air inhaled after a normal inspiration 40) _______
41) the minimum amount of air always present in the respiratory system, after blowing out all you can 41) _______
42) the extra amount actively (forcibly) exhaled after a normal exhalation 42) _______
43) the amount of air taken in during a single normal inspiration 43) _______

A) inspiratory reserve volume
B) residual volume
C) expiratory reserve volume
D) tidal volume

Match the lung capacity with its description.

44) the amount of air remaining in the lungs after a normal breath 44) _______

A) functional residual capacity

45) the sum of all the lung volumes — 45) _______

46) the amount of air inhaled during an active (forced) inspiration — 46) _______

47) the total amount of air that can be exchanged at will — 47) _______

A) total lung capacity

B) vital capacity

C) inspiratory capacity

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

48) The beating of the cilia of the respiratory passages in the direction of the pharynx forms a ________. 48) _______________

49) When the inspiratory muscles relax, the rib cage returns to its original position as a result of ________. 49) _______________

50) The ease with which the lungs stretch in response to changes in pressure is termed ________. 50) _______________

51) The ability of a lung to recoil, or recover from stretch, is called ________. 51) _______________

52) In the disease ________, many symptoms are due to destruction of elastic fibers in the lung. 52) _______________

53) Some have a congenital alpha-1 antitrypsin deficiency. If the job of alpha-1 antitrypsin is to inhibit trypsin and elastase, what condition are people with this genetic deficiency likely to develop? 53) _______________

54) The substance produced by the lungs to reduce surface tension is called ________. 54) _______________

55) The cells of the lung that produce the substance that lowers surface tension are ________. 55) _______________

56) Because of their smooth muscle component, the structures of the lower respiratory system that can most alter airway resistance are the ________. 56) _______________

57) A powerful bronchoconstrictor released by mast cells is ________. 57) _______________

58) Ongoing diseases in which air flow during expiration is diminished are known as ________. 58) _______________

59) An increase in the rate and volume of breathing is known as ________. 59) _______________

60) The volume of air moved in a single respiration is called the ________. 60) _______________

61) The volume of air that can be forcefully expelled from the lungs following a normal exhalation is called the ________. 61) _______________

62) The volume of air that can be forcefully inhaled after a normal inspiration is called the ________. 62) ______________

63) The volume of air that remains in the lungs after a maximal expiration is called the ________. 63) ______________

64) A typical residual volume (in milliliters) for a healthy, 70 kg male is about ________. 64) ______________

65) The ________ consist of light, spongy tissue whose volume is occupied mostly by air-filled spaces. 65) ______________

66) The opposing layers of pleural membrane are held together by a thin film of ________. 66) ______________

67) The larynx contains the ________, connective tissue bands that tighten and vibrate to create sound when air moves past them. 67) ______________

68) The primary function of the alveoli is ________. 68) ______________

69) The pressure contributed by a single gas in a mixture of gases is known as ________. 69) ______________

70) A ________ is an instrument that measures the volume of air moved with each breath. 70) ______________

71) Air flows into lungs because of ________ created by ________. 71) ______________

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

72) List the muscles of inspiration and expiration.

73) Define lung capacity, and give a specific example. (Note: There is a specific physiological definition; don't just define "capacity" in general terms.)

74) List and describe the lung volumes.

75) Explain how the upper airways and bronchi condition the air, and why this conditioning is important.

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

76) During normal expiration, 76) _______

A) the abdominal muscles become involved.
B) elastic recoil of stretched muscles helps return the thorax to its resting volume, the internal intercostal muscles are required, and the abdominal muscles become involved.
C) elastic recoil of stretched muscles helps return the thorax to its resting volume and the internal intercostal muscles are required.
D) the internal intercostal muscles are required.
E) elastic recoil of stretched muscles helps return the thorax to its resting volume.

77) Damage to the type II cells of the lungs would contribute to 77) _______
A) alveolar collapse.
B) decreased surface tension in the water lining the alveoli.
C) alveolar rupture.
D) an increased rate of gas exchange.
E) a thickening of the respiratory membrane.

78) Harry suffers from cystic fibrosis and frequently has periods where he can hardly breathe. The problem is the result of 78) _______
A) laryngospasms that occur in response to a toxic substance produced by the epithelial cells.
B) inflammation of the bronchi.
C) constriction of the trachea.
D) thick secretions that exceed the ability of the mucus elevator to transport them.
E) collapse of one or both lungs.

79) Breathing that involves active inspiratory and expiratory movements is called ________ because it ensures that a person can increase overall volume. 79) _______
A) hyperpnea.
B) costal breathing.
C) eupnea.
D) diaphragmatic breathing.
E) shallow breathing.

80) The respiratory rate times the tidal volume corrected for dead space is the 80) _______
A) respiratory minute volume.
B) pulmonary ventilation rate.
C) alveolar ventilation rate.
D) vital capacity.
E) external respiration rate.

81) Increasing the alveolar ventilation rate will 81) _______
A) decrease the rate of carbon dioxide diffusion from the blood to the alveoli.
B) decrease the rate of oxygen diffusion from the alveoli to the blood.
C) increase the partial pressure of oxygen in the alveoli.
D) increase the partial pressure of carbon dioxide in the alveoli.
E) have no effect on either the partial pressure or diffusion rate of gases.

82) In the lungs, an example of the body's reserve capacity is that 82) _______
A) pulmonary blood flow is completely under the control of the autonomic nervous system, dilating arteries and arterioles to adjust blood flow.
B) capillary beds in the lungs are reversibly collapsible, allowing blood to be shunted to additional areas during exercise.
C) some areas of the lung can be closed off during rest and opened again when needed during exercise.
D) All of the statements are true.
E) None of the statements are true.

83) Joe is playing in an intramural football game when he is tackled so hard that he breaks a rib. He can actually feel a piece of the rib sticking through the skin, and he is having a difficult time breathing. Joe probably is suffering from 83) _______
A) an obstruction in the bronchi.
B) a collapsed trachea.
C) a pneumothorax.
D) decreased surfactant production.
E) a bruised diaphragm.

84) In a condition known as pleurisy, there is excess fluid in the pleural space. How would you expect this to affect the process of pulmonary ventilation? 84) _______
A) More air would be forced out during expiration.
B) Tidal volume would increase.
C) Breathing would be labored and difficult.
D) Ventilation would require less energy.
E) It would be easier to expand the lungs on inspiration.

MATCHING. Choose the item in column 2 that best matches each item in column 1.

Match the type of breathing with its description.

85) cessation of breathing	A) apnea	85) _______
86) increased respiratory rate and/or volume without increased metabolism	B) hyperventilation	86) _______
	C) dyspnea	
87) increased respiratory rate and/or volume due to increased metabolism	D) tachypnea	87) _______
88) rapid breathing	E) hyperpnea	88) _______
89) difficulty breathing		89) _______

Match the change in gas composition with the response (disregard weak responses).

90) Arteries: systemic constrict, pulmonary dilate.	A) P_{CO_2} decreases.	90) _______
	B) P_{O_2} increases.	
91) Bronchioles and systemic arteries dilate.	C) P_{CO_2} increases.	91) _______
92) Arteries: systemic dilate, pulmonary constrict.	D) P_{O_2} decreases.	92) _______
93) Bronchioles and systemic arteries constrict.		93) _______

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

94) Which single muscle contributes most to normal, resting inspiration?

95) Determine the difference between cellular and external respiration.

96) Describe the pleura and explain its role in respiration.

97) Distinguish between intrapulmonary and intrapleural pressure.

98) What is surfactant? Why is it important?

99) Premature babies frequently need to be put on a respirator to help them breathe. Why does this become a necessary treatment for many premature babies?

100) What is Dalton's law? Why is it important?

101) Air flow in the respiratory system and blood flow in the cardiovascular system have many similarities–name them. How are they different?

102) What are the functions of pleural fluid?

103) Summarize the four gas laws.

104) During the winter, Ali sleeps in a dorm room that lacks a humidifier for the heated air. In the mornings he notices that his nose is "stuffy," similar to when he has a cold, but after showering and drinking some water, the stuffiness disappears until the next morning. What might be the cause of Brad's nasal condition?

105) A newborn infant is found dead, abandoned by the road. Among the many questions that the police would like to have answers to is whether the infant was born dead or alive. After an autopsy, the medical examiner tells them that the infant was dead at birth. How could the medical examiner determine this?

106) Ralph is taking SCUBA diving classes and is confused as to why he should not hold his breath under water while ascending–his instructor told him he must exhale continuously. What would you tell him?

107) Tenzin is not able to produce surfactant. In order to inhale a normal tidal volume, does her intrapleural pressure have to be higher or lower than for a healthy individual? Explain.

108) A package insert for a medication states that "this medicine is a beta receptor stimulant used to treat symptoms of asthma, emphysema and other breathing conditions. Possible side effects include fast heart rate, nervousness, tremors or nausea." Why would these symptoms be expected?

109) You were the top student in your physiology class last semester. For this reason, your professor has asked that yo prepare some class sessions for her physiology class this semester. Specifically, she would like you to discuss the similarities and differences between the cardiovascular and respiratory systems, in terms of pressure gradients a flow.

A. Explain how you will describe the role of the following structures: the pump and the tubes.

B. Explain how you will describe the importance of an open vs. closed system and the relevant differences betwe liquids and gases.

C. What is fundamentally different about the respiratory pump compared to the cardiac pump, and why does th difference exist?

110) How would a spirometer tracing be different in a person with a collapsed lung? Why? What must occur to restore the respiratory function to normal?

111) With what you have learned about pulmonary ventilation, how would you design an artificial respirator to keep patients with paralyzed respiratory muscles alive? How does artificial respiration by a machine compare to mouth–to–mouth respiration?

112) Ariana "gets the wind knocked out of her" during a skiing accident in which she attempted a jump and landed hard on her feet before tumbling over. She is conscious but breathing is labored, and she complains of pain and shortness of breath. The first aid–trained ski patrol person that comes to her aid determines that Ariana has no broken bones, nor is CNS injury likely. He places tiny tubes at her nostrils that blow air (with higher than normal percentage of oxygen) into her nose then places her on the snowmobile, and takes her to the first aid tent. Why is Ariana's breathing labored? How does the high–oxygen air help her condition?

113) Nami is breathing 14 times per minute, with a tidal volume of 520 mL and a dead space of 152 mL. Lyle is breathing 15 times per minute, with a tidal volume of 400 mL and a dead space of 175 mL. Which patient has better alveolar ventilation?

114) Weiru is taking 14 breaths per minute. Her vital capacity is 3000 mL, her total lung capacity is 4000 mL, and her t volume is 450 mL per breath. Calculate the following:
A. Weiru's total pulmonary ventilation (minute volume)
B. Weiru's alveolar ventilation rate

115) There is a mixture of gases in dry air, with an atmospheric pressure of 760 mm Hg. Calculate the partial pressure gas for the examples below:
A. 20.7% oxygen, 78.2% nitrogen, 0.4% carbon dioxide
B. 45% oxygen, 41% nitrogen, 3% carbon dioxide, 11% hydrogen
C. 79% oxygen, 15% nitrogen, 5% carbon dioxide, 1% argon

116) Draw and label a spirogram showing respiratory activity in a resting person. Draw and label a spirogram in this same person during exercise.

117) Draw three graphs showing how the following change during normal inspiration and expiration:
A. alveolar pressure
B. intrapleural pressure
C. volume of air moved

118) A. Omar is snorkeling (surface swimming with a tube in his mouth to maintain breathing while his face is in the The snorkel tube is 40 cm in length and 1.2 cm in radius. How does this affect Omar's breathing? Calculate the in relative resistance contributed by the snorkel tube.
B. What is dead space and how does that affect snorkeling? Explain your answer, assuming the snorkel tube is a cylinder and cylinder volume is pr^2L. How does the snorkel tube affect alveolar ventilation (assume a ventilatio of 12 breaths/min, tidal volume of 500 mL, and normal dead space volume of 150 mL)? If normal alveolar ventila 4200 mL/min, how must Omar breathe during snorkeling compared to normal?
C. Omar loves snorkeling, but wishes he could go a little deeper. He decides to create a longer snorkel tube to try achieve this desire. He isn't sure how radius figures in, so he makes two tubes of 60 cm length, one with a 1.2 cm radius and the other with a 2 cm radius. What are the relative resistances of these two new tubes? Do you think he will be able to snorkel well with either of these tubes? Explain.

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

119) Alexandra is breathing in a gas composition that contains an increased concentration of CO_2. How would the pulmonary arteries react to this and why? 119) ______

A) Dilate: increase blood flow through the alveoli
B) Dilate: decrease blood flow through the alveoli
C) Constrict: increase blood flow through the alveoli
D) Constrict: decrease blood flow through the alveoli

Answer Key
Testname: UNTITLED36

1) D
2) B
3) C
4) A
5) D
6) A
7) C
8) B
9) A
10) B
11) B
12) C
13) C
14) B
15) C
16) D
17) A
18) A
19) B
20) D
21) A
22) B
23) D
24) E
25) E
26) B
27) B
28) C
29) C
30) B
31) B
32) D
33) E
34) D
35) A
36) C
37) C
38) D
39) A
40) A
41) B
42) C
43) D
44) A
45) A
46) C
47) B
48) mucociliary escalator
49) elastic recoil
50) compliance

51) elastance
52) emphysema
53) emphysema
54) surfactant
55) type II alveolar cells
56) bronchioles
57) histamine
58) obstructive lung diseases
59) hyperventilation
60) tidal volume
61) expiratory reserve volume
62) inspiratory reserve volume
63) residual volume
64) 1200 mL
65) lungs
66) pleural fluid
67) vocal cords
68) the exchange of gases between themselves and the blood
69) partial pressure
70) spirometer
71) pressure gradients, a pump
72) Inspiration: diaphragm, external intercostals, scalenes, and sternocleidomastoids.
Expiration: internal intercostals, abdominal wall muscles.
73) A lung capacity is the sum of two or more lung volumes. The lung capacities are total lung capacity, vital capacity, functional residual capacity, and inspiratory capacity.
74) residual volume (quantity of air that cannot be expelled), expiratory reserve volume (quantity of air that is expelled only with forced expiration), tidal volume (quantity of air that is inhaled and exhaled under a specific condition, usually normal resting breathing), inspiratory reserve volume (quantity of air that can be inhaled with a forceful inspiration)
75) Air is warmed to body temperature to protect alveoli and avoid disrupting body temperature, air is humidified so that the exchange epithelium will not dehydrate, and foreign material is filtered out so it will not reach the alveoli.
76) E
77) A
78) D
79) A
80) C
81) C
82) B
83) C
84) C
85) A
86) B
87) E
88) D
89) C
90) B
91) C
92) D
93) A

94) The diaphragm muscle provides most of the force for inspiration. Normal expiration is primarily a result of the diaphragm relaxing, and expiratory muscles do not contribute.
95) Cellular respiration refers to the intracellular reactions that consume oxygen and produce ATP. External respiration refers to the exchange of gases between the environment and the lungs, lungs and blood, blood and cells, and the transport of those gases.
96) The pleura is a double-walled sac that surrounds each lung. One wall clings to the lung surface, the other to the wall of the thoracic cavity. The pleura secretes and contains pleural fluid, which is a lubricant. Pleural fluid also keeps the wall of the lung in close proximity to the wall of the thoracic cavity, which is critical in the process of pulmonary ventilation.
97) Intrapulmonary pressure is the pressure inside the alveoli. Intrapleural pressure is the pressure within the pleural cavity. Intrapleural pressure variation drives variation inside the alveoli, but is always lower than atmospheric pressure, where pressure inside the alveoli equilibrates with atmospheric pressure during the respiratory cycle.
98) Surfactant, made by the type II alveolar cells, reduces the surface tension in the fluid in the alveoli, thereby facilitating inflation and inhibiting collapse of the alveoli.
99) The lungs are one of the last organs to develop during pregnancy and the type II alveolar cells are particularly late in developing. Since the type II alveolar cells secrete surfactant, premature babies may not make enough surfactant to reduce the resistance to breathing and may need positive pressure oxygen to be given along with surfactant to help them breathe on their own.
100) Dalton's law states that the total air pressure in a mixture of gases is the sum of the pressures contributed by each individual gases (the partial pressures). It is important because the air that we breathe is a mixture of gases, gas pressure is related to amount of gas, and the respiratory system is regulated in part by the partial pressures of oxygen and carbon dioxide in the body.
101) 1. Flow takes place from regions of higher pressure to regions of lower pressure.
2. A muscular pump creates pressure gradients.
3. Resistance to fluid flow is influenced primarily by the diameter of the tubes.
The primary difference is that air is a compressible mixture of gases while blood is a non-compressible liquid.
102) 1. It creates a moist, slippery surface so opposing membranes can slide across one another as the lungs move within the
2. It holds the lungs tight against the thoracic wall.
103) 1. The total pressure of a mixture of gases is the sum of the pressures of the individual gases (Dalton's law).
2. Gases, singly or in a mixture, move from areas of higher pressure to areas of lower pressure.
3. If the volume of a container of gas changes, the pressure of the gas will change in an inverse manner (Boyle's law).
4. The pressure and volume of a container of gas are directly related to the temperature of the gas and the number of molecules in the container: $PV = nRT$ (the ideal gas law).
104) Since the air that Ali is breathing is not humidified (thus dry), large amounts of moisture are leaving the mucus to humidify the air that is being respired. This makes the mucus tacky and the cilia have difficulty moving. As more mucus is produced, it builds up, forming the nasal congestion in the morning. As Brad showers and drinks fluid, the moisture is replaced and the mucus loosens up and is moved along the proper route as usual. The reason this happens mostly at night is because Brad is probably not getting up frequently to drink water to replace what is being lost to humidify the air.
105) Unless the infant was suffocated immediately at birth, the first breath that it took would start to inflate the lungs and some of the air would be trapped in the lungs. By placing the lungs in water to see if they float or not, the medical examiner can determine whether or not there is any air in the lungs. Other measurements and tests could also be used to determine if the infant had breathed at all (air in the lungs) or was dead at birth (lungs collapsed and a small amount of fluid in the alveoli).
106) When breathing the compressed air from a SCUBA tank, the lungs are virtually completely inflated. Problems would arise if he held his breath and began to ascend. On ascending, the decrease in pressure would cause the gas in his lungs to expand (Boyle's law). Since the lungs are already fully inflated, this would cause over-inflation and possible rupturing of the lung. If this occurred, air would leak from the lungs into the pleural cavity, resulting in a pneumothorax and a collapsed lung.

107) Surfactant is necessary to reduce surface tension sufficient to prevent small or collapsed alveoli. Sherry needs to inhale more forcefully to get the same tidal volume, so lower intrapleural pressure would help.

108) Stimulation of beta receptors activates the sympathetic division, overriding the parasympathetic division. Increased heart rate, nervousness, nausea, and tremors all accompany sympathetic stimulation.

109) A. In both systems, fluids flow down a pressure gradient, within different-sized tubes of variable resistance; the pressure gradient is generated by the action of a muscular pump. In the cardiovascular system, the tubes vary from elastic arteries that minimize pressure extremes and fluctuations, muscular arteries and arterioles whose diameter is changeable to regulate flow and that account for most of the resistance, finally to the large number of tiny capillaries in parallel, with veins to conduct the blood back to the pump. In the respiratory system, the larger tubes are rigid and acco most of the resistance, and lead to a large number of tiny tubes in parallel whose diameter is adjusted to regulate flow.
B. In the closed cardiovascular system, the highest pressure is generated within the left ventricle, and an incompressible flows in a continuous fashion. High pressure is created when the ventricle contracts, squeezing the blood it contains; lov pressure results when the ventricle relaxes. In the open respiratory system, the highest pressure during normal breathing is generated when the respiratory muscles relax, decreasing the volume of the alveoli as air flows out; low pressure results when the muscles contract. Air is compressible, thus Boyle's law of inverse relationship between volum and pressure is important. The flow of air is intermittent, flowing into the lungs then stopping, and flowing out of the lu stopping again.
C. The driving pressure for fluid flow in the cardiovascular system is created directly by contraction of the heart chamb depends upon the total peripheral resistance. The driving pressure for fluid flow in the respiratory system is created indirectly by contraction of respiratory muscles, which produces a volume increase that creates a pressure decrease relative to atmospheric pressure. The compressibility of air and the incompressibility of blood, as well as the fact that the respiratory system is open and thus dependent on atmospheric pressure, while the cardiovascular system is closed and thus wholly responsible for pressure, explains these differences.

110) Each volume would be roughly reduced by half, because only one lung is functioning. The collapsed lung cannot be used for pulmonary ventilation until the damage to the pleura is repaired and the low intrapleural pressure and residual volume is reestablished.

111) The artificial respirator needs to alternate the pressure it generates between slightly higher than atmospheric pressure and slightly lower than atmospheric pressure. These pressure fluctuations are what the respiratory system normally produces; thus they should be safe for the respiratory structures. The lungs of the patient will passively inflate and deflate in response, and gas exchange at the alveoli will occur. The principle is the same with mouth-to-mouth respiration.

112) Ariana suffered an impact that increased her intrapleural pressure above normal, possibly by pushing her diaphragm upward, forcing her to expel some of her residual volume. With her alveoli smaller than normal, they are very difficult to inflate. The lack of broken ribs suggests that she does not have a pneumothorax. The air being blown into her nostrils has a higher P_{O_2}, which will increase the diffusion into her blood and compensate for her temporarily decreased tidal volume.

113) Alveolar ventilation = ventilation rate × (tidal volume - dead space volume). Julia's ventilation is 5432 mL/min, which is better than Lyle's, which is 3375 mL/min.

114) A. 14 breaths/min × 450 mL/breath = 6300 mL/min
B. Alveolar vent = 14 breaths/min × (450 mL/ breath - 150 mL dead space) = 14 breaths/min × 300 mL/breath = 4200 mL/min

115) Partial pressure = P_{atm} × % gas in atmosphere.
A. 157.32 mm Hg oxygen, 594.32 mm Hg nitrogen, 3.04 mm Hg carbon dioxide
B. 342 mm Hg oxygen, 311.6 mm Hg nitrogen, 22.8 mm Hg carbon dioxide, 83.6 mm Hg hydrogen
C. 600.4 mm Hg oxygen, 114 mm Hg nitrogen, 38 mm Hg carbon dioxide, 7.6 mm Hg argon.

116) The resting graph should resemble Figure 17.7. During exercise, the only things that change are the size of the tidal volume curve, which will become larger as it enters into the inspiratory and expiratory reserve capacities, and these reserve capacities will have the same extreme values but will be smaller.

117) See Figure 17.9 in the chapter.

118) A. The snorkel tube effectively increases the total length of the respiratory passage, which increases the resistance to air flow ($R \propto L\eta/r^4$); this makes Omar work harder to breathe. Resistance increases proportionally to L/r^4; $40/1.2^4 = 19$.
B. Dead space is the volume of air that enters the respiratory structures, but does not reach the alveoli and cannot contri gas exchange. The snorkel tube increases the amount of dead space by p $1.2^2 \times 40 = 181$ mL, which more than doubles t dead space to 331 mL. Alveolar ventilation = ventilation rate × (tidal volume − dead space) = 12/min × (500 mL − 331 mL) = 2028 mL/min; this is less than half of normal alveolar ventilation. To maintain 4200 mL/min, Omar could increase his ventilation rate to 25 breaths/min (4200/(500 − 331)). Alternatively, he could increase his ventilation depth to 681 mL/breath (4200/12 + 331). Or he could alter both rate and depth.
C. Tube 1: $60/1.2^4 = 29$. Tube 2: $60/2^4 = 3.75$. The first tube would require significantly more effort to use than the tube i A, so he would not be able to snorkel as long, if at all. The second tube sounds desirable, but dead space must be consid The snorkel tube adds an additional
p 2 $cm^2 \times 60$ cm = 754 mL of dead space, which is greater than normal tidal volume and would reduce alveolar ventilation. Neither tube is practical.
119) D

Exam

Name________________________________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) High carbon dioxide concentration in body fluids is called 1) ________
A) hyperoxia.
B) carbonation.
C) hypercapnia.
D) hypercarbia.
E) hyperdioxia.

2) The partial pressure of oxygen in arterial blood is approximately ________ mm Hg. 2) ________
A) 100 B) 45 C) 50 D) 70 E) 40

3) The partial pressure of carbon dioxide in the cells of peripheral tissues is approximately ________ mm Hg. 3) ________
A) 40 B) 100 C) 50 D) 46 E) 70

4) Of the factors that influence diffusion of respiratory gases, the most variable and, therefore, important factor to consider is the 4) ________
A) membrane surface area.
B) concentration gradient.
C) electrical charge.
D) membrane thickness.
E) diffusion distance.

5) The process by which dissolved gases are exchanged between the blood and interstitial fluids is 5) ________
A) pulmonary ventilation.
B) breathing.
C) diffusion.
D) cellular respiration.
E) external respiration.

6) The lung pathology most likely to result from certain kinds of heart disease is 6) ________
A) asthma.
B) emphysema.
C) pulmonary edema.
D) lung cancer.
E) fibrotic lung disease.

7) Hypoxia resulting from fluid accumulation in the alveoli that cannot be corrected by oxygen therapy can lead to 7) ________
A) sudden infant death syndrome.
B) emphysema.
C) asthma.
D) adult respiratory distress syndrome.
E) fibrotic lung disease.

8) If the partial pressure of oxygen in both air and water is 100 mm Hg, then the concentration of the oxygen is the same in the air and water. 8) ________
A) True
B) False

9) The variables of oxygen consumption, cardiac output, and blood oxygen content are unrelated. 9) ________
A) True
B) False

10) Which characteristic makes hemoglobin's structure such a good match for its function as an oxygen carrier? 10) ________
A) Each hemoglobin molecule can bind one oxygen molecule.
B) Each hemoglobin molecule cannot be saturated by oxygen molecules.
C) Each hemoglobin molecule can bind two oxygen molecules.
D) Each hemoglobin molecule can bind four oxygen molecules.
E) Each hemoglobin binds irreversibly to an oxygen molecule.

11) Most of the oxygen transported by the blood is 11) ________
A) carried by white blood cells.
B) bound to hemoglobin.
C) bound to a plasma protein.
D) in ionic form as solute in the plasma.
E) dissolved in plasma.

12) At a P_{O_2} of 70 mm Hg and normal temperature and pH, hemoglobin is ________% saturated with oxygen. 12) ________
A) 25
B) 10
C) 50
D) 75
E) over 90

13) Which would make the oxygen-hemoglobin curve shift right? 13) ________
A) increased H^+ concentration
B) decreased CO_2
C) increased pH
D) decreased temperature
E) None of the answers are correct.

14) Chronic hypoxia 14) ________
A) can be caused by anemia.
B) increases 2,3-BPG production in blood.
C) shifts the HbO_2 dissociation curve to the left.
D) increases 2,3-BPG production in blood and can result from anemia.
E) All of the answers are correct.

15) Most of the carbon dioxide in the blood is transported as 15) ________
A) solute dissolved in the cytoplasm of red blood cells.
B) carbonic acid.
C) carbaminohemoglobin.
D) bicarbonate ions.
E) solute dissolved in the plasma.

16) In the medulla oblongata, the nucleus tractus solitarius contains the ________ of neurons. 16) ________
A) dorsal respiratory group
B) pontine respiratory group
C) pre-Botzinger complex
D) ventral respiratory group

17) The most important chemical regulator of respiration is 17) _______
A) carbon dioxide.
B) oxygen.
C) bicarbonate ion.
D) sodium ion.
E) hemoglobin.

18) An increase in the level of carbon dioxide in the blood will 18) _______
A) increase the rate of breathing.
B) decrease the alveolar ventilation rate.
C) increase the pH of arterial blood.
D) decrease pulmonary ventilation.
E) decrease the rate of breathing.

19) The expiratory neurons control the ________ muscles, whereas the inspiratory neurons control the ________ muscles. 19) _______
A) abdominal muscles and internal intercostals, diaphragm and external intercostals
B) diaphragm and abdominal muscles, internal and external intercostals
C) diaphragm and internal intercostals, abdominal muscles and external intercostals
D) diaphragm and external intercostals, abdominal muscles and internal intercostals
E) abdominal muscles and external intercostals, diaphragm and internal intercostals

20) If the neural connections between the pons and medulla are cut, breathing will stop. 20) _______
A) True B) False

21) The Hering–Breuer reflex 21) _______
A) alters pulmonary ventilation when the P_{O_2} changes.
B) is an important aspect of normal, quiet breathing.
C) limiting ventilation volumes.
D) alters pulmonary ventilation when the P_{CO_2} changes.
E) functions to increase ventilation with changes in blood pressure.

22) Protective reflexes of the lungs include 22) _______
A) bronchodilation and coughing.
B) bronchodilation.
C) coughing.
D) bronchoconstriction.
E) coughing and bronchoconstriction.

MATCHING. Choose the item in column 2 that best matches each item in column 1.

Match the lung disease to its description.

23) destruction of alveoli — 23) _______

24) thickened alveolar membrane and decreased lung compliance — 24) _______

25) decreased surface area for gas exchange — 25) _______

26) increased airway resistance — 26) _______

27) fluid accumulation in interstitial spaces — 27) _______

28) increased diffusion distance — 28) _______

A) pulmonary edema

B) asthma

C) emphysema

D) fibrotic lung disease

Match the factor with its effect on the affinity of hemoglobin for oxygen.

29) increased temperature — 29) _______

30) increased pH — 30) _______

31) increased P_{CO_2} — 31) _______

32) increased 2,3–BPG — 32) _______

A) increase

B) decrease

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

33) Too little oxygen in cells is called ________. — 33) _______________

34) Too little oxygen in cells is often accompanied by too much ________, which is called ________. — 34) _______________

35) Generally, P_{O_2} in arterial blood is ________ than P_{O_2} in venous blood. — 35) _______________

36) Generally, P_{CO_2} in arterial blood is ________ than P_{CO_2} in venous blood. — 36) _______________

37) P_{CO_2} tends to be ________ in tissues than in systemic capillaries. — 37) _______________

38) Diffusion rate is directly proportional to ________, ________ and ________. — 38) _______________

39) ________ is characterized by a decreased surface area for gas exchange in the lungs. — 39) _______________

40) ________ is characterized by a thickened alveolar membrane, slowing respiratory gas exchange. 40) ______________

41) In ________, fluid accumulates in the interstitial spaces of the lungs, slowing gas exchange. 41) ______________

42) ________ is characterized by an increased airway resistance and decreased ventilation. 42) ______________

43) Diffusion rate is indirectly proportional to ________. 43) ______________

44) ________ is the enzyme that converts CO_2 into bicarbonate ions. 44) ______________

45) The ________ group of neurons contains mostly inspiratory neurons. The ________ group of neurons controls muscles used for active expiration and some inspiratory muscles. 45) ______________

46) The output of the ________ group of inspiratory neurons controls the ________ muscle(s) by way of the ________ nerve. 46) ______________

47) Inappropriate relaxation of the ________ muscles during sleep contributes to ________, a sleep disorder associated with snoring. 47) ______________

48) Specialized ________ in the carotid and aortic bodies are activated by a decrease in P_{O_2} and pH or an increase in P_{CO_2}. What do they trigger? 48) ______________

49) The carotid and aortic bodies contain specialized ________ cells, which can increase ventilation in response to changes in P_{O_2}, P_{CO_2}, or pH. 49) ______________

50) Fear and excitement may affect the pace and depth of respiration by stimulation of portions of the ________. 50) ______________

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

51) What force(s) move(s) carbon dioxide from the blood into the alveoli?

52) List and explain the factors that influence gas diffusion in the lungs.

53) Name three pathological changes that adversely affect gas exchange.

54) List and explain the factors that influence the presence of gases in liquids.

55) What are the three ways CO_2 is transported in blood? Approximately what percentage is transported by each way?

56) What are the P_{O_2} and P_{CO_2} in the alveoli, pulmonary artery, peripheral tissue, and systemic veins? Explain why the P_{O_2} and P_{CO_2} change.

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

57) Mara lives in St. Louis, which is close to sea level. She decides to spend a month of her summer vacation working in the mountains outside of Denver. After a week in the mountains, what kinds of changes would you expect to see as Mara adapts to the higher altitude? 57) _______

A) decreased P_{O_2} in the alveoli
B) decreased alveolar ventilation rate
C) decreased hematocrit
D) decreased blood pressure
E) All of the answers are correct.

58) Carbon dioxide is more soluble in water than oxygen. To get the same amount of oxygen to dissolve in plasma as carbon dioxide, you would have to 58) _______

A) decrease the partial pressure of nitrogen.
B) increase the rate of plasma flow through the lungs.
C) increase the partial pressure of oxygen.
D) decrease the alveolar ventilation rate.
E) decrease the temperature of the plasma.

59) For maximum efficiency in loading oxygen at the lungs, 59) _______

A) the P_{O_2} should be about 70 mm Hg.
B) BPG levels in the red blood cells should be high.
C) the pH should be slightly acidic.
D) the temperature should be slightly lower than normal body temperature.
E) All of the answers are correct.

60) A student in your lab volunteers to enter a hypoxic breathing chamber for 10 minutes, and his alveolar P_{O_2} drops to 50 mm Hg. What other change would occur? 60) _______

A) decrease in arterial pH
B) decrease in pH of cerebrospinal fluid
C) decrease in arterial P_{CO_2}
D) hypoventilation
E) increase in alveolar P_{CO_2}

61) A inhibitor of carbonic anhydrase would 61) _______

A) increase the amount of bicarbonate formed in the blood.
B) increase blood pH.
C) interfere with oxygen binding to hemoglobin.
D) decrease the amount of carbon dioxide dissolved in plasma.
E) All of the answers are correct.

62) The chloride shift occurs when 62) _______

A) hydrogen ions leave the red blood cells.
B) carbonic acid is formed.
C) bicarbonate ions leave the red blood cells.
D) hydrogen ions enter the red blood cells.

63) Blocking afferent action potentials from the chemoreceptors in the carotid and aortic bodies would interfere with the brain's ability to regulate breathing in response to all EXCEPT which changes? 63) _______

A) changes in pH due to carbon dioxide levels
B) changes in P_{O_2}
C) changes in P_{CO_2}
D) changes in blood pressure
E) All of the answers are correct.

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

64) Describe the different causes of hypoxia, and give specific examples of the associated conditions.

65) What are the two possible causes of lower alveolar P_{O_2}? Give examples of each.

66) Describe the problems that result in low arterial oxygen content.

67) Compare and contrast the pulmonary pathologies that affect alveolar ventilation and gas exchange.

68) Explain how oxygen and carbon dioxide are transported in the blood. How does the means of transport relate to the solubility and chemical reactivity of these gases in plasma?

69) List and describe the brain's centers for monitoring and controlling respiration.

70) List and describe the locations and stimuli for respiratory chemoreceptors.

71) Draw a flow chart that shows the steps of the reflex loop in which an increase in blood P_{CO_2} leads to increased ventilation.

72) A 10–year–old swimmer has attached an extension to his snorkel tube, so that he can sit on the bottom of the ocean for a longer period of time, watching the fish of the coral reef. What problems would you expect him to have? Assuming he is successful in breathing this way for a little while, what are some of the differences he will experience compared to normal breathing?

73) Drowning often occurs when outside water enters the alveoli, yet some water is normally present in alveoli. Why is water normally present in the alveoli? What is present in alveolar fluid, besides water, that aids in respiration? What properties of water are harmful to respiration? What causes drowning? While extremely rare, there have even been documented cases of people walking away from recreational swimming only to die later, on dry land, as a result of drowning (termed "delayed" or "secondary" drowning). Propose an explanation for delayed drowning.

74) In the science fiction movie The Abyss, a diver is able to breathe while his head is in a specially prepared liquid pumped into a water–tight helmet attached to a suit, thus allowing him to dive without an air tank and at greater depths than possible for scuba divers. Is this purely fiction, or is it at least theoretically possible? Explain, thinking about the characteristics of alveoli and alveolar gas exchange. Assuming it is possible, describe the characteristics of the liquid in terms of its oxygen pressure and concentration.

75) Do the factors affecting affinity of hemoglobin for oxygen have the same effect on loading and unloading oxygen in lungs and tissues? Explain your answer, and indicate if this presents a conflict in responding to hypoxia. Assuming a medical team has treatments available for changing all the factors affecting hemoglobin's oxygen affinity, how should they change pH, temperature, P_{CO_2}, and 2,3–BPG content in a hypoxic patient?

76) Hemoglobin binds to and has a much higher affinity for carbon monoxide (CO) than for oxygen. CO is colorless and odorless and can be produced in homes heated by natural gas; victims usually do not realize they are being poisoned and often die in their sleep. Describe the likely changes in a CO poisoning victim.

77) Compare and contrast carbon monoxide (CO) poisoning, in which CO displaces oxygen from hemoglobin, and metabolic poisoning such as by cyanide.

78) Write the chemical equation catalyzed by the enzyme carbonic anhydrase. Suppose the concentration of H^+ is increased by an outside force in a solution that had been at equilibrium. According to the law of mass action, what must happen to the CO_2 concentration to reestablish equilibrium after this disturbance? What must happen to the concentration of bicarbonate?

79) A chemistry student accidentally spills chlorine bleach into a dilute acid. The mixture reacts and produces fumes that are inhaled by the student and that reduce his ventilation. Assume that the gaseous chemical produced is a base (i.e., it releases OH^- in an aqueous solution), and that the chemical is absorbed into the bloodstream at the alveoli. Explain why ventilation is reduced in the patient.

80) You are a scientist who has been hired to write a screenplay for a popular TV series. The plot involves the discovery by a botanist of a plant toxin that interferes with the function of the exchange pump responsible for the chloride shift. This scientist provides the toxin to bioterrorists. What should you write for the coroner to say about symptoms in the poisoning victims?

81) Which is usually more important in regulating the respiratory system, P_{O_2} or P_{CO_2}? Explain your answer and briefly discuss the receptors involved. Give examples of situations in which each of those factors changes enough to stimulate a reflex. How and why are these factors related to each other?

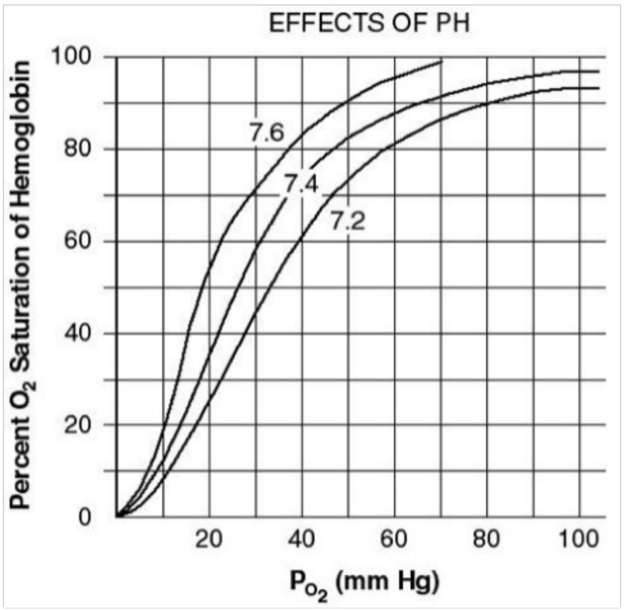
82) Timmy is a toddler who has just threatened that he will hold his breath until his mother gives him some chocolate. His mother refuses to be manipulated and watches as Timmy stubbornly refuses to breathe. To her horror, Timmy loses consciousness and collapses onto the floor. Her cousin, who is enrolled in a course for emergency medical technical (EMT) training, is visiting, and tells her there is no need to call for an ambulance. Why did Timmy lose consciousness? Should his mother trust her instincts and call for help and begin CPR, or should she listen to her well–meaning but young and inexperienced cousin? Explain.

83) Cary deliberately hyperventilates for several minutes before diving into a swimming pool. Shortly after he enters the water and begins swimming, he blacks out and almost drowns. What caused this to happen?

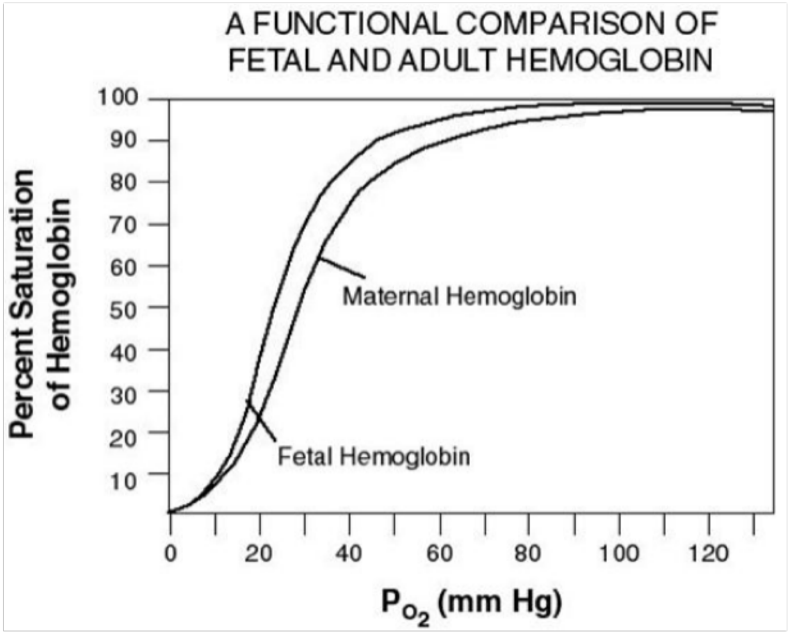
84) Define hyperventilation and explain what may cause it. Is the increased ventilation that occurs while exercising an example of hyperventilation? Explain your answer. How are P_{O_2} and P_{CO_2} affected by hyperventilation? Can breathing into a paper bag decrease hyperventilation? Explain.

85) Oxygen consumption increases with exercise due to the fact that the contracting muscles are doing work and req ATP to produce the contractions.
A. How does the body meet the increased demand for oxygen?
B. Calculate the oxygen consumption of a runner using the following information:
Heart Rate: 130 bpm; Stroke Volume: 270 mL/beat;
Arterial oxygen content: 250 mL O_2/L blood, and
Venous oxygen content: 100 O_2/L blood.

86) The graph below shows an oxygen dissociation curve, with the normal curve in the center. What is the shift in th to the right called? What causes this change in oxygen dissociation? Is the shift helpful or harmful? Explain.



87) Refer to the graph below.



A. At a P_{O_2} of 80 mm Hg, which type of hemoglobin binds more oxygen?

B. At a P_{O_2} of 40 mm Hg, which type of hemoglobin has released more oxygen to the cell?

C. Explain the significance of the differences in fetal and maternal hemoglobin affinity.

D. If a worm lived in low oxygen mud flats where the P_{O_2} is 60 mm Hg, which type of hemoglobin would be be for it to have? Explain.

Answer Key
Testname: UNTITLED37

1) C
2) A
3) D
4) B
5) C
6) C
7) D
8) B
9) B
10) D
11) B
12) E
13) A
14) D
15) D
16) A
17) A
18) A
19) A
20) B
21) C
22) E
23) C
24) D
25) C
26) B
27) A
28) A
29) B
30) A
31) B
32) B
33) hypoxia
34) carbon dioxide, hypercapnia
35) higher
36) lower
37) higher
38) the available surface area, the concentration gradient of the gas, the permeability of the barrier
39) Emphysema
40) Fibrotic lung disease
41) pulmonary edema
42) Asthma
43) square of the distance (membrane thickness)
44) Carbonic anhydrase
45) dorsal respiratory, ventral respiratory
46) dorsal respiratory, diaphragm (or intercostals), phrenic (or intercostal)
47) mouth and throat (larynx, pharynx, and tongue), obstructive sleep apnea
48) glomus cells
They trigger a reflex increase in ventilation.
49) glomus

50) limbic system
51) Carbon dioxide moves into the alveoli as a result of the pressure gradient.
52) Concentration gradient, surface area, membrane thickness, and diffusion distance are the four factors. Increasing the concentration gradient and surface area will increase the rate of diffusion across the lungs while increasing membrane thickness and diffusion distance will decrease the rate of diffusion in the lungs. This is discussed in the "Diffusion and Solubility of Gases" section of the chapter.
53) 1. a decrease in the amount of alveolar surface area available for gas exchange
2. an increase in the thickness of the alveolar membrane
3. an increase in the diffusion distance between the alveoli and the blood
54) Pressure, solubility, and temperature are three factors. This is discussed in "Gas Solubility Affects Diffusion" section of the chapter.
55) 1. attached to hemoglobin, 23%
2. dissolved, 7%
3. as bicarbonate, 70%
56) See Figure 18.2 and the "Gas Exchange in the Lungs and Tissues" section of the chapter.
57) A
58) C
59) D
60) C
61) B
62) C
63) D
64) See Table 18.1 in the chapter.
65) 1. The composition of the inspired air is abnormal. Altitude affects oxygen content of air.
2. Alveolar ventilation is inadequate. Pathological factors include increased airway resistance (asthma), decreased lung compliance (fibrosis), and overdoses of drugs or alcohol.
66) Three categories of problems are inadequacies in oxygen reaching alveoli, oxygen exchange between the alveoli and blood, and transport of oxygen in the blood.
67) See Figure 18.3(c) in the chapter, which discusses emphysema, fibrotic lung disease, pulmonary edema, and asthma.
68) Oxygen is not highly soluble in water, which is the main component of plasma. Less than 2% is dissolved in plasma, with the remainder bound to hemoglobin. The iron in the heme portion of the molecule can bind up to four oxygen atoms. Oxygen is not chemically reactive in the body. Carbon dioxide is more soluble than oxygen, at about 7% dissolved. Carbon dioxide is chemically reactive, combining with water to form carbonic acid, which then dissociates to bicarbonate and hydrogen ion. Most carbon dioxide is transported in the form of bicarbonate, about 70%, and the remaining 23% binds to amino acids on hemoglobin.
69) Respiratory neurons in the medulla control inspiration and expiration. Neurons in the pons modulate ventilation. The rhythmic pattern of breathing arises from a network of spontaneously discharging neurons. Ventilation is subject to modulation by various chemical factors and by higher brain centers.
70) Peripheral chemoreceptors in the carotid and aortic bodies sense changes in oxygen concentration, pH, and P_{CO_2} of the plasma. Central chemoreceptors monitor cerebrospinal fluid composition and respond to changes in the concentration of CO_2 in the cerebrospinal fluid.
71) Peripheral chemoreceptors in carotid and aortic bodies detect changes in P_{CO_2} and pH. Central chemoreceptors in the medulla oblongata monitor CO_2 in CSF. Either an increase in P_{CO_2} or decrease in pH will stimulate the receptors, which project to a control center in the medulla oblongata. The control center stimulates somatic motor neurons that control the skeletal muscles involved in ventilation. The effect is increased ventilation, which lowers P_{CO_2} by eliminating CO_2, so blood pH increases because of this shift. See Figure 18.17 in the chapter.

72) The resistance to air flow increases as a result of adding the extension to the snorkel tube. Assuming he has used an extension of sufficiently large diameter, he is in a shallow location where the total length isn't excessive, and he has the strength and endurance in his respiratory muscles, he may be able to breathe this way for a few seconds longer than he could have held his breath. He will probably notice that he is breathing more deeply and at a higher rate, to maintain ventilation against the higher resistance. After a few exchanges he will start to hypoventilate, despite his efforts to increase pulmonary ventilation, because the presence of dead airspace in the tube will result in decreasing P_{O_2} in the alveoli. Dead airspace is a problem if the total volume of air in the tube is greater than the volume exchanged with each breath.
73) Water is present everywhere in living tissues, because all living cells require water as a diffusion medium for solutes. Alveolar fluid is present in only a thin layer because it contains surfactant, preventing it from collapsing the alveoli due to surface tension. Oxygen has low solubility in water, but the large total surface area of alveoli combined with the thinness of the water layer in the alveoli allows for sufficient oxygen to enter the blood. Inhalation of water increases the amount of water and decreases the relative amount of surfactant. Less oxygen will reach the alveolar membranes, and alveoli will collapse. Alveolar collapse combined with the low solubility of oxygen in water will lead to hypoxemia and will trigger cardiovascular and respiratory reflexes. Delayed drowning likely results from tissue damage in the lungs and malfunction in other organs resulting from hypoxemia and reflexes related to the water inhalation.
74) (Note to instructor: Such experiments were done in the 1960s with rodents, and were partially successful but ultimately deemed impractical and too risky for humans.) It is theoretically possible. Oxygen dissolves in a thin film of fluid in the alveoli, thus even in normal breathing a liquid containing dissolved oxygen is involved. Obviously the respiratory gases would have to be soluble in this liquid and in concentrations appropriate to maintain the partial pressures necessary to drive diffusion.
75) Hemoglobin affinity changes in the same way in the lungs and tissues, thus increasing unloading in the tissues does decrease loading in the lungs, which seems as if one cancels out the other. But from the oxygen dissociation curves it is evident that the impact is greater in the tissues; thus increased unloading can remedy hypoxia. The medical team should decrease pH, increase temperature, increase P_{CO_2}, and increase 2,3-BPG.
76) CO will gradually displace O_2 on hemoglobin molecules. While this increases O_2 unloading in tissues, it will significantly decrease O_2 loading in lungs, therefore hypoxia will result. Victims gradually lose consciousness as the brain tissues become hypoxic.
77) Both types of poisoning interfere with oxygen-dependent metabolism and thus can be fatal. CO prevents oxygen loading at the lungs and thus produces hypoxia. Metabolic poisons exert their effects on the chemical reactions that consume oxygen to produce ATP, and thus have no effect on P_{O_2}.
78) $CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$

Carbon dioxide concentration will increase as the reaction is shifted to the left. Bicarbonate concentration will decrease.
79) The physiological response is opposite that for acidosis. A decreased ventilation would reduce loss of CO_2. By maintaining higher P_{CO_2} in the blood, the resulting acidity would help to counteract the alkalinity resulting from the inhalation of a base.
80) The chloride shift is the transport process that occurs in red blood cells and is necessary for normal carbon dioxide transport. In this process, a bicarbonate ion is exchanged for a chloride ion. As a result, the red blood cell loses bicarbonate, which carries a negative charge, but gains a chloride ion, thus maintaining its membrane potential. Losing the bicarbonate prevents the chemical reaction that produces bicarbonate from carbon dioxide and water from reaching equilibrium. Bicarbonate is the most important extracellular buffer in the body. Without the transporter function, bicarbonate will build up inside the red blood cell and will not be maintained in the plasma. With a reduction in plasma bicarbonate, acidosis will result. An increase in pulmonary ventilation is triggered by acidosis and can help reverse this process, but it may be unable to compensate fully for the lack of bicarbonate buffer. Chapter 20 covers some of the physiological consequences of acidosis.

81) P_{CO_2} is the more important factor. For P_{CO_2}, there are central and peripheral receptors that respond to CO_2 as well as to CO_2-related pH. These receptors are very sensitive to routine changes in P_{CO_2} and pH, such as those associated with an increase in physical activity. Peripheral chemoreceptors have been identified for O_2, but these respond only to dramatic changes in P_{O_2}, such as those associated with high altitude or disease. Because CO_2 is produced as a by-product of aerobic (oxygen-consuming) metabolism, an increase in CO_2 is associated with a corresponding decrease in O_2.

82) Timmy loses consciousness due to hypoxia in his brain. There is no need for alarm, because the loss of consciousness indicates mainly that cerebral activity has decreased. The parts of the brain involved in respiratory control are in the brain stem. As long as Timmy was awake and determined, his cerebral signals were able to inhibit brain stem control of pulmonary ventilation. Once he loses consciousness, however, the respiratory control areas are released from inhibition and he starts breathing again. Soon normal oxygen content in the brain will be restored, and Timmy will wake up.

83) Hyperventilation causes a decrease in the alveolar P_{CO_2}, and more carbon dioxide is eliminated from the body than during normal breathing. The loss of large amounts of carbon dioxide upsets the body's normal drive for ventilation, and Cary does not feel the urge to breathe as he swims. As the exercising muscles use oxygen, a state of hypoxia develops. This results in insufficient amounts of oxygen reaching the brain, causing Cary to lose consciousness.

84) Hyperventilation is an increase in alveolar ventilation that exceeds metabolic demand. A person can deliberately hyperventilate, or it may occur as a result of emotional stress or high altitude. Strictly speaking, the increase in ventilation during exercise is necessary to meet increased metabolism and is therefore not hyperventilation. P_{O_2} is increased and P_{CO_2} is decreased during hyperventilation, because more O_2 is inhaled and more CO_2 is exhaled; this produces abnormally low P_{CO_2}. Hyperventilation can be remedied by a paper bag treatment, because rebreathing exhaled air will increase P_{CO_2} in the body back to normal.

85) In order to meet the increased oxygen usage, more blood needs to be delivered to the working muscles. This is accompli increasing cardiac output and by vasodilation of blood vessels to the exercising muscles thus increasing flow and delive more oxygen rich blood. Oxygen consumption can be calculated using the Fick equation.
Q_{O_2} = CO × (arterial oxygen content − venous oxygen content)
However, CO must be calculated first: CO = SV × HR = 130 × 250 = 32.5 L/min
Then Q_{O_2} = 32.5 × (0.250 − 0.100) = 4.875 L O_2/min

86) The shift in the dissociation curve is called the Bohr effect. The decrease in pH may be due to an increase in CO_2 and lactic acid during vigorous physical activity, or to metabolic acidosis. The Bohr effect is helpful, because the unloading of oxygen from hemoglobin increases under such conditions. If the decrease in pH is due to exercise, then the skeletal muscles have an increased demand for oxygen, and thus increased unloading is desirable.

87) A. Fetal hemoglobin has a higher affinity for oxygen than maternal hemoglobin at that oxygen pressure.
B. Maternal hemoglobin releases oxygen more readily than fetal hemoglobin at that oxygen pressure.
C. The higher affinity of fetal hemoglobin for oxygen ensures a net transfer of oxygen from the maternal blood to the fet blood in the placenta. While unloading from fetal hemoglobin to fetal tissues is less favorable due to the higher affinity, rate of oxygen consumption in fetal tissues and consequent low oxygen pressure ensures unloading will occur.
D. The fetal form of hemoglobin would be better in this situation, because it has a higher affinity for the oxygen at this p Thus, the worm's blood would become more saturated, and more oxygen would then be available for the worm's cells.

Exam

Name________________________________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) Which is NOT a function of the kidneys? 1) ________
A) regulation of extracellular fluid volume
B) maintenance of ion balance in body fluids
C) homeostatic regulation of blood pH
D) regulation of blood osmolarity
E) regulation of blood protein levels

2) Which ion is NOT directly regulated by the kidney? 2) ________
A) Ca^{2+} B) HCO_3^- C) K^+ D) OH^- E) Na^+

3) The characteristic yellow color of urine is from the presence of 3) ________
A) urobilinogen.
B) bile.
C) uric acid.
D) urea.
E) renin.

4) Urine is produced by the 4) ________
A) gallbladder.
B) urethra.
C) kidney.
D) ureter.
E) urinary bladder.

5) Urine is carried to the urinary bladder by 5) ________
A) lymphatics.
B) the ureters.
C) the urethra.
D) blood vessels.
E) All of the answers are correct.

6) Technically, the kidneys are located 6) ________
A) in the pelvic cavity.
B) in the abdominal cavity.
C) behind the peritoneal membrane.
D) behind the pleural membranes.
E) in the thoracic cavity.

7) Blood flow through the kidney includes a feature seen in only a few organs. What is it? 7) ________
A) arterial shunts
B) portal system
C) vascular sinuses
D) veins containing highly oxygenated blood
E) anastomoses

8) Which structure is NOT part of the blood circulation through the kidney? 8) ________
A) glomerulus B) renal corpuscle C) vasa recta D) loop of Henle

9) Which statement is FALSE? 9) ________
A) The urinary bladder is filled by two ducts, called ureters, and emptied by the single urethra.
B) Women are more likely to develop urinary tract infections than men.
C) The kidneys do not reabsorb filtered waste products.
D) The blood supply to the kidneys comes through the renal arteries.
E) As much as one fourth of the cardiac output may flow to the kidneys at any given moment.

10) The Bowman's capsule and glomerulus make up the 10) ________
A) renal papilla.
B) renal corpuscle.
C) loop of Henle.
D) renal pyramid.
E) collecting system.

11) Which kidney process is always active and always requires energy to occur? 11) ________
A) filtration B) secretion C) reabsorption D) excretion

12) A glomerulus is 12) ________
A) the expanded end of a nephron.
B) a "ball" of capillaries within Bowman's capsule.
C) the hairpin–shaped segment of the nephron.
D) the portion of the nephron closest to the renal corpuscle.
E) the portion of the nephron that attaches to the collecting duct.

13) The portion of the nephron closest to the renal corpuscle is the 13) ________
A) distal tubule.
B) minor calyx.
C) collecting duct.
D) proximal tubule.
E) loop of Henle.

14) The portion of the nephron that attaches to the collecting duct is the 14) ________
A) distal tubule.
B) collecting duct.
C) proximal tubule.
D) minor calyx.
E) loop of Henle.

15) The hairpin–shaped segment of the nephron is the 15) ________
A) distal tubule.
B) minor calyx.
C) proximal tubule.
D) loop of Henle.
E) vasa recta.

16) The segment of the nephron between the proximal and distal tubules that loops down into the medulla of the kidney and returns back to the cortex is called the 16) _______
A) collecting duct.
B) minor calyx.
C) Bowman's capsule.
D) loop of Henle.
E) vasa recta.

17) The process of filtration in the kidney is most accurately described as 17) _______
A) relatively nonspecific. B) highly specific. C) completely nonspecific.

18) Which is NOT normally be found in the filtrate? 18) _______
A) erythrocytes. B) potassium. C) urobilinogen. D) glucose.

19) Cysts on the kidney can press upon nephrons, increasing the pressure inside the nephrons. How will this affect glomerular filtration rate and blood pressure? 19) _______
A) GFR increases and blood pressure decreases
B) GFR decreases and blood pressure increases
C) GFR increases and blood pressure increases
D) GFR decreases and blood pressure decreases

20) The amount of plasma that filters into the nephrons is approximately ________ of the total volume. 20) _______
A) 1/5 B) 90% C) 4/5 D) 3/4 E) 1/2

21) In normal kidneys, blood cells and plasma proteins are 21) _______
A) filtered and secreted. B) filtered then reabsorbed.
C) secreted then reabsorbed. D) not filtered.

22) Which is NOT a kidney filtration barrier? 22) _______
A) juxtaglomerular apparatus B) basement membrane
C) glomerular capillary endothelium D) Bowman's capsule epithelium

23) The force for glomerular filtration is the 23) _______
A) fluid pressure produced by the displacement of the fluid in the lumen of the tubules.
B) ATP-dependent processes in the nephron.
C) blood pressure in the glomerular capillaries.
D) osmotic pressure in the glomerular capillaries.
E) None of the answers are correct.

24) The primary function of the proximal tubule is 24) _______
A) secretion of acids and ammonia.
B) adjusting the urine volume.
C) secretion of drugs.
D) reabsorption of ions, organic molecules, and water.
E) filtration.

25) Contents in the peritubular capillaries are actively transported into proximal and distal convoluted tubules in a process known as 25) _______
A) secretion. B) filtration. C) excretion. D) reabsorption.

26) Glucose and amino acids are reabsorbed by — 26) _______
A) cotransport.
B) countertransport.
C) symport with sodium.
D) diffusion.

27) Which statement about autoregulation is TRUE? — 27) _______
A) In myogenic response, the macula densa cells send a paracrine message to the neighboring afferent arteriole.
B) Myogenic response is the intrinsic ability of vascular smooth muscle to respond to pressure changes.
C) Myogenic response is a paracrine signaling mechanism.
D) In tubuloglomerular feedback, stretch-sensitive ion channels open, resulting in depolarization of smooth muscle cells.

MATCHING. Choose the item in column 2 that best matches each item in column 1.

Match each step in urine formation to its description.

28) movement from the nephron lumen to the external environment — 28) _______
29) movement from the nephron lumen to the blood — 29) _______
30) movement from the glomerulus to the nephron lumen — 30) _______
31) movement from the peritubular capillaries to the nephron lumen — 31) _______

A) secretion
B) excretion
C) reabsorption
D) filtration

Match each substance with its primary mode of transport across the kidney epithelium.

32) sodium — 32) _______
33) glucose — 33) _______
34) urea — 34) _______
35) small plasma proteins — 35) _______

A) symport with a cation
B) passive reabsorption/diffusion
C) transcytosis
D) active transport

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

36) After it is formed, urine is temporarily stored in the ________. — 36) _______________

37) Urine is carried to the external environment by the ________. — 37) _______________

38) The plasma concentration at which all of the renal carriers for a given substance are saturated is the ________. — 38) _______________

39) The amount of filtrate entering Bowman's capsule each minute is the ________. — 39) _______________

40) The percentage of total plasma volume that filters is called the ________. 40) ______________

41) When fluid flow through the distal tubule increases as a result of increased GFR, the macula densa cells send a chemical message to the neighboring afferent arteriole. The afferent arteriole constricts, increasing resistance and decreasing GFR. This type of autoregulation involving both the kidney tubule and the arteriole is known as ________. 41) ______________

42) The ________ branch off the ________ and supply blood to the kidneys. 42) ______________

43) The ________ carry blood from the kidneys back to the ________. 43) ______________

44) Eighty percent of the nephrons in a kidney are contained within the ________, but the other 20%, called the ________ nephrons, dip down into the ________. 44) ______________

45) The ________ are the long peritubular capillaries that dip into the medulla. 45) ______________

46) The nephron begins with a hollow, ball-like structure called ________. 46) ______________

47) The ________ lie between and around the glomerular capillaries. 47) ______________

48) The specialized cells found in the capsule epithelium are called ________. These cells have long cytoplasmic extensions called ________. 48) ______________

49) Neural control of GFR is mediated by ________ that innervate ________ receptors on vascular smooth muscle causing ________. 49) ______________

50) The excretion of glucose in the urine is called ________. 50) ______________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

51) Place the following blood vessels that carry blood to and within the kidney in the order in which blc passes through them. 51) _______

1. afferent arteriole
2. efferent arteriole
3. glomerulus
4. peritubular capillary

A) 1, 3, 2, 4 B) 4, 3, 2, 1 C) 1, 2, 3, 4 D) 4, 2, 3, 1

52) When the plasma concentration of a substance exceeds its renal concentration, more of the substance will be 52) _______

A) secreted.
B) filtered.
C) excreted.
D) reabsorbed.
E) None of the answers are correct.

53) Damage to the renal medulla would interfere first with the functioning of the 53) ________
A) Bowman's capsule.
B) glomerulus.
C) collecting ducts.
D) distal tubule.
E) proximal tubule.

54) An obstruction in a glomerulus would affect the flow of blood into the 54) ________
A) renal vein.
B) renal artery.
C) afferent arteriole.
D) efferent arteriole.

55) If blood flow through the afferent arterioles increases, 55) ________
A) stretch stimulates vasoconstriction to reduce the flow.
B) the smooth muscle in the vessel walls stretches to accommodate the increased flow and the stretch stimulates further relaxation of the arteriolar wall, lessening blood pressure.
C) the smooth muscle in the vessel walls stretches to accommodate the increased flow.
D) the stretch stimulates further relaxation of the arteriolar wall, lessening blood pressure.

56) Urea is 56) ________
A) completely eliminated in the urine.
B) actively transported into the filtrate by the cells of the collecting duct.
C) actively secreted in the distal tubule.
D) passively reabsorbed in the proximal tubule.
E) actively reabsorbed in the proximal tubule.

57) In the lumen of the proximal tubule, Na^+ concentration is ________ the Na^+ concentration inside the cells of the tubule wall. 57) ________
A) much higher than
B) much lower than
C) slightly lower than
D) slightly higher than
E) about the same as

58) The basic pattern for many molecules absorbed by Na^+–dependent transport involves this: an apical ________ and a basolateral ________. 58) ________
A) symport protein, osmotic gradient
B) facilitated diffusion carrier, osmotic gradient
C) facilitated diffusion carrier, symport protein
D) symport protein, facilitated diffusion carrier
E) osmotic gradient, symport protein

59) One substance has no membrane transporters to move it but can diffuse freely through open leak channels if there is a concentration gradient. Initially, this substance's concentrations in the filtrate and extracellular fluid are equal. Later, however, the active transport of Na^+ and other solutes creates a gradient by removing water from the lumen of the tubule where it is located. What substance is this? 59) _______

A) glucose
B) urea
C) calcium
D) glucose and calcium
E) glucose, calcium, and urea

60) Measurements in a nephron reveal a glomerular hydrostatic pressure of 69 mm Hg, and a fluid pressure in the Bowman's capsule of 15 mm Hg. Assuming that the plasma colloid osmotic pressure is 30 mm Hg, and that essentially no plasma proteins are filtered by the glomerulus, what is the net glomerular filtration pressure in this case? 60) _______

A) 114 mm Hg
B) 84 mm Hg
C) 54 mm Hg
D) -6 mm Hg
E) 24 mm Hg

61) In a normal kidney, which condition would increase glomerular filtration rate (GFR)? 61) _______

A) a decrease in the net glomerular filtration pressure
B) a decrease in the hydrostatic pressure of the glomerulus
C) an increase in the hydrostatic pressure in Bowman's capsule
D) constriction of the afferent arteriole
E) a decrease in the concentration of plasma proteins in the blood

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

62) A person with cirrhosis of the liver has lower than normal levels of plasma proteins and a higher than normal GFR. Explain why a decrease in plasma protein concentration would cause an increase in GFR.

63) Patrick's urine sample reveals a high concentration of glucose. Is glucose normally present in urine? Suggest two possible mechanisms to explain why the kidney would excrete excess glucose, and what abnormality may underlie those conditions.

64) Trace a drop of water through the urinary system, beginning with the plasma in the renal artery.

65) List and describe the three filtration barriers that substances leaving the plasma must pass through before entering the tubule lumen.

66) List and summarize the forces that influence filtration across the walls of glomerular capillaries.

67) Reabsorption involves which two methods of transport? Describe each. What determines which route a solute will take?

68) Describe GFR autoregulation, and explain why it is important.

69) Ruby is 77 years old and can no longer control her urination. What changed in the reflex pathway that used to control her urination?

70) Sylvia is suffering from severe edema in her arms and legs. Her physician prescribes a diuretic (a substance that will increase the volume of urine produced). Why might this help to alleviate Sylvia's problem?

71) Make a map of the following terms: *blood, Bowman's capsule, collecting duct, distal tubule, excretion, external environment, filtration, loop of Henle, lumen, osmosis, peritubular capillaries, proximal tubule, urine, water.* Add terms as needed.

72) A professor gave the following analogy of kidney function to his freshman biology class: "Imagine your desk is a disorganized mess, containing things worth keeping and things you should throw out. Instead of picking through each item one by one and considering if it is to keep or throw out, you sweep your arm across your desktop in one smooth motion, knocking the entire contents of the desk into your large trash can. Then you pull the items you want to keep out of the trash can and place them back on your desk." In what ways is this analogy accurate? In what ways is it inaccurate? Some students in his class argue that the kidneys are inefficient and poorly designed. How should the professor respond? What modern medical device may he cite in his response?

73) Presence of glucose in the urine is evidence of what condition?

74) Armann weighs 100 kg. Assume his total blood volume is 7%, his heart pumps his total blood volume within 90 seconds, and his renal blood flow is 20% of his cardiac output. Calculate the volume of blood that flows through Armann's kidneys each minute.

75) You are a physiologist on a space flight to a distant planet. You discover intelligent human–like aliens on the planet. You discover that the alien's kidney handles the sugar alcohol mannitol just like our kidneys handle inuli also test the alien for its renal handling of glucose. The data from that experiment are: Plasma level (constant ove hrs): 6 g/dL of mannitol and 2 g/dL of glucose; 24 hr. urine sample: volume of 2 L, containing 144 g mannitol and glucose.
A. What is the alien's glomerular filtration rate?
B. What is the alien's clearance rate for glucose?
C. How does the alien kidney handle glucose?

76) A man's leg was crushed between a car bumper and a wall. His physicians believe their patient has suffered kid damage from myoglobin blocking glomerular pores. Tests showed the following results:
plasma creatinine: 30 mg/100 mL plasma
24 hour urine specimen: volume = 1 liter
urine creatinine: 30 mg/mL urine

A. How many mg of creatinine are in the urine specimen? How much creatinine appears in the urine per hour?
B. What is the patient's creatinine clearance in mL/min?
C. What is the patient's GFR?
D. Is this a normal GFR? Did the patient sustain kidney damage?

Answer Key
Testname: UNTITLED38

1) E
2) D
3) A
4) C
5) B
6) C
7) B
8) D
9) C
10) B
11) B
12) B
13) D
14) A
15) D
16) D
17) A
18) A
19) B
20) A
21) D
22) A
23) C
24) D
25) A
26) C
27) B
28) B
29) C
30) D
31) A
32) D
33) A
34) B
35) C
36) urinary bladder
37) urethra
38) renal threshold
39) GFR or glomerular filtration rate
40) filtration fraction
41) tubuloglomerular feedback
42) renal arteries, abdominal aorta
43) renal veins, inferior vena cava
44) cortex, juxtamedullary, medulla
45) vasa recta
46) Bowman's capsule
47) mesangial cells
48) podocytes, foot processes
49) sympathetic neurons, alpha (or α), vasoconstriction
50) glucosuria or glycosuria

51) A
52) C
53) C
54) D
55) A
56) D
57) A
58) D
59) B
60) E
61) E
62) The primary driving force for GFR is blood pressure, opposed by fluid pressure in Bowman's capsule and osmotic pressure due to plasma proteins. If a person has fewer plasma proteins due to liver disease, the plasma will have a lower osmotic pressure. With less osmotic pressure opposing the GFR, GFR will increase.
63) Glucose transporters may be saturated, preventing complete reabsorption of glucose (filtration exceeds reabsorption). This can happen with extremely high glucose ingestion or with untreated diabetes mellitus. A genetic defect resulting in an insufficient number of glucose transporters is another possibility.
64) Plasma in the renal artery flows into smaller branches, entering the arcuate arteries, which lead to the afferent arterioles, which in turn lead to the glomeruli. Water moves from glomeruli into the nephrons (PCT, loop of Henle, then DCT), into the collecting ducts, then renal calyces and into the renal pelvis, passes into a ureter, then to the urinary bladder; finally, it is expelled as urine through the urethra.
65) 1. the glomerular capillary endothelium
2. a basal lamina
3. the epithelium of Bowman's capsule
See the "The Renal Corpuscle Contains Filtration Barriers" section of the chapter.
66) 1. hydrostatic pressure
2. colloid osmotic pressure
3. hydrostatic fluid pressure
See the "Capillary Pressure Causes Filtration" section of the chapter.
67) 1. transepithelial transport: substances cross both apical and basolateral membranes of the tubule epithelial cell.
2. paracellular pathway: substances pass through the junction between two adjacent cells.
The permeability of the epithelial junction and the electrochemical gradient for the solute determines which route it wil
68) 1. The myogenic response is the ability of the vascular smooth muscle to constrict or dilate to keep pressure and flow co thereby keeping GFR constant. Vasoconstriction is more effective at this than vasodilation.
2. Tubuloglomerular feedback involves production of a paracrine, in response to increased GFR, which constricts the af arteriole to decrease GFR. Autoregulation helps protect the filtration barriers from high blood pressures that could dam them.
69) Micturition normally occurs when stretch receptors in the bladder wall are stimulated by distension due to the accumulation of urine. The stretch receptors project to an integration center in the spinal cord, which stimulates parasympathetic neurons of the bladder wall, which stimulates bladder contraction, forcing the internal urethral sphincter to open. At the same time, the spinal control center inhibits somatic motor neurons that control the external urethral sphincter, which relaxes in response, so that urine can flow out. The spinal reflex is normally overridden by descending control from the brain stem and cerebral cortex. The brain inhibits the parasympathetic output to the bladder and stimulates the somatic motor output to the external urethral sphincter, thus preventing urination until a convenient time. It is likely that Ruby's descending control is what is no longer working properly; thus she is not able to suppress the urge until she can get to a toilet.

70) One simple answer is that a diuretic reduces the free fluid in the body. A more complex explanation is that increasing the volume of urine produced would decrease the total blood volume of the body. This in turn would lead to a decreased blood hydrostatic pressure. Edema is frequently the result of hydrostatic pressure of the blood exceeding the opposing forces at the capillaries in the affected area. Depending on the actual cause of the edema, decreasing the blood hydrostatic pressure would decrease edema formation and possibly cause some of the fluid to move from the interstitial space back to the blood. Diuresis would also increase the concentration of the proteins in the plasma, contributing to the fluid's movement out of the tissues and into the blood.
71) Maps will vary. See Figure 19.2 and the "Overview of Kidney Function" section of the chapter.
72) His analogy is accurate in that the glomerulus does dump contents into the nephron that the body needs and will take back. His analogy is inaccurate in that the entire contents of the plasma are not dumped, rather about 1/5 of the volume enters the nephron. While one could easily imagine a more efficient way to remove waste from the blood, the kidneys can only use the tools they have, namely various membrane transport processes and the physical laws governing movement of water and solutes. Kidney dialysis machines rely on diffusion through semipermeable membranes, mimicking part of normal kidney function.
73) Diabetes mellitus
74) Armann's blood volume = 100 kg × 0.07 = 7 L = 7000 mL. At least two thirds of this volume must circulate per minute = 7000 mL × 0.67 = 4690 mL/min minimum cardiac output (CO); values up to 5000 mL/min would still be within a normal range; 4690 × 20% = 938 mL/min; 5000 × 20% = 1000 mL/min, thus blood flow to Arman's kidneys approximates 938–1000 ml/min.
75) Human kidneys filter inulin, a polysaccharide from dahlia roots, but do not reabsorb or secrete it. Thus, 100% of the inu urine sample is filtered inulin. For this reason, the inulin excretion rate is the same as the glomerular filtration rate.
A. The alien's GFR = excretion rate of mannitol/plasma concentration of mannitol
= 144 g/day × 1 dL/6 g × 0.1 L/1 dL = 2.4 L/day.
B. Using 2.4 L/day as GFR, the filtered load for glucose = plasma concentration of glucose × GFR
= 2 g/dL × 2.4 L/day × 1 dL/0.1 L = 48 g glucose.
C. There were 52 g of glucose produced in 24 hours. Since the clearance of glucose is greater than the filtered load, some must have been secreted. Therefore the alien's kidney filters and secretes glucose.
76) A. 1000 mL urine × 30 mg creatinine/mL urine = 30,000 mg creatinine in the specimen.
30,000 mg/24 hr = 1250 mg creatinine.
B. Creatinine clearance = excretion rate of creatinine/plasma concentration creatinine.
clearance = 1250 mg creatinine/hr × 1 hr/60 min/30 mg creatinine/100 mL plasma
= 20.8 mg creatinine/min/0.3 mg creatinine/mL plasma
= 69.4 mL plasma/min
C. GFR approximately equal to clearance rate for creatinine, or approximately 70 mL/min.
D. A more typical value for GFR is 125 mL/min, so 70 mL/min represents a severe decline. Thus, kidney damage has mo occurred.

Exam

Name________________________________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) The primary route for water loss from the body is the ________ system. 1) ________
A) respiratory
B) digestive
C) cardiovascular
D) integumentary
E) urinary

2) The primary route for ion loss from the body is the ________ system. 2) ________
A) urinary
B) integumentary
C) digestive
D) respiratory
E) cardiovascular

3) Cell volume (and therefore cell function) in most cells is dependent upon careful regulation of 3) ________
A) osmolarity of extracellular fluid.
B) volume of extracellular fluid.
C) permeability of cell membranes.
D) blood pressure.
E) resting membrane potential.

4) The two organ systems that work together to regulate most aspects of the body's water balance are 4) ________
A) urinary and cardiovascular.
B) urinary and respiratory.
C) digestive and cardiovascular.
D) digestive and respiratory.
E) cardiovascular and respiratory.

5) Kidneys respond relatively ________ to changes in blood volume. 5) ________
A) quickly
B) slowly

6) Shrinkage of hepatocytes in the liver results in which of the following? 6) ________
A) protein synthesis only
B) glycogen production only
C) glycogen breakdown only
D) both glycogen production and protein synthesis
E) both glycogen breakdown and protein breakdown

7) Most body water is located in 7) ________
A) lumens of organs open to the outside.
B) cells.
C) plasma.
D) interstitial fluid.

8) Kidneys regulate 8) ________
A) water gain only.
B) water loss only.
C) both water loss and gain.

9) When a body is dehydrated, water in the urinary bladder 9) ________
A) can be returned to the circulation after moving back into the kidneys.
B) will still be expelled from the body in the urine.
C) can be returned to the circulation directly.

10) Water reabsorption by the kidneys is a result of 10) _______
A) exchange with ions.
B) both passive and active transport processes.
C) osmosis.
D) cotransport with ions.

11) The hormone that directly controls water reabsorption by the kidneys is 11) _______
A) angiotensin.
B) vasopressin.
C) aldosterone.
D) ANP.
E) epinephrine.

12) The hormone that regulates water reabsorption by the kidneys 12) _______
A) only increases water permeability in certain portions of the kidney tubules.
B) decreases water permeability throughout the kidney tubules.
C) only decreases water permeability in certain portions of the kidney tubules.
D) increases water permeability throughout the kidney tubules.

13) Why do patients taking loop diuretics need to take supplemental potassium? 13) _______
A) They inhibit the reabsorption of potassium as well as sodium in the loop of Henle.
B) They cause active reabsorption of potassium in the distal convoluted tubule.
C) They inhibit intestinal absorption of potassium.
D) They cause active secretion of potassium in the loop of Henle.

14) The primary osmoreceptors are located in the 14) _______
A) medulla.
B) hypothalamus.
C) pons.
D) stomach.
E) kidney.

15) Osmoreceptors fire after they ________ in response to ________ plasma osmolarity. 15) _______
A) swell, increased
B) shrink, decreased
C) shrink, increased
D) swell, decreased

16) The hormone vasopressin 16) _______
A) stimulates the kidneys to retain water.
B) stimulates the kidneys to produce a large volume of urine.
C) stimulates the kidneys to retain sodium ions.
D) is secreted by the anterior pituitary gland in response to changes in blood osmolarity.
E) All of the answers are correct.

17) When venous return is increased, stretch receptors in the atria of the heart are activated. This results in 17) _______
A) increased vasopressin secretion.
B) increased thirst.
C) decreased urine production.
D) decreased vasopressin secretion.
E) increased glomerular filtration.

18) When baroreceptors in the carotid and aortic bodies sense increased blood pressure, this results in 18) _______
A) increased thirst.
B) increased glomerular filtration.
C) decreased urine production.
D) decreased vasopressin secretion.
E) increased vasopressin secretion.

19) Why is sodium actively reabsorbed in the nephron? 19) _______
A) to increase passive reabsorption of water
B) to decrease blood pressure
C) to make urine less concentrated
D) to decrease osmolarity inside the nephron

20) Granular cells secrete 20) _______
A) aldosterone.
B) renin.
C) angiotensinogen.
D) angiotensin converting enzyme.
E) angiotensin I.

21) ACE converts 21) _______
A) angiotensinogen to angiotensin I.
B) angiotensin II to aldosterone.
C) renin to aldosterone.
D) renin to angiotensinogen.
E) angiotensin I to angiotensin II.

22) Stimuli for the activation of the RAS pathway include 22) _______
A) low blood pressure in arterioles in the nephron only.
B) low blood pressure in arterioles in the nephron and a decrease in fluid flow through the distal tubule.
C) a decrease in fluid flow through the distal tubule only.
D) high blood pressure in the renal artery only.
E) low blood pressure in arterioles in the nephron, a decrease in fluid flow through the distal tubule, and high blood pressure in the renal artery.

23) Angiotensin II stimulates 23) _______
A) thirst, vasoconstriction, and release of aldosterone.
B) thirst only.
C) vasoconstriction only.
D) release of aldosterone.
E) thirst and vasoconstriction.

24) Which is FALSE about angiotensin II? 24) _______
A) stimulates thirst
B) stimulates vasoconstriction
C) activates parasympathetic output
D) increases blood pressure
E) increases cardiac output

25) Drugs that treat hypertension by preventing Angiotensin I from becoming Angiotensin II are called 25) _______
A) ACE inhibitors.
B) calcium channel blockers.
C) beta blockers.
D) diuretics.

26) Aldosterone 26) _______
A) functions in pH regulation.
B) helps decrease blood volume.
C) stimulates sodium reabsorption in kidney principal cells.
D) increases the concentration of sodium in urine.
E) is secreted in response to increased levels of sodium in the blood.

27) Atrial natriuretic peptide 27) _______
A) inhibits release of renin.
B) increases GFR and inhibits release of renin.
C) increases GFR and stimulates release of renin.
D) increases GFR.
E) stimulates release of renin.

28) An increase in plasma potassium levels is properly called 28) _______
A) hypernatremia.
B) hyperkalemia.
C) hypercalcemia.
D) hyperpotassemia.
E) hyperpotasseplasmia.

29) Excess potassium ions are eliminated from the body by the 29) _______
A) sweat glands.
B) liver.
C) spleen.
D) kidneys.
E) digestive system.

30) Thirst is 30) ________
A) controlled by centers in the hypothalamus, stimulated by increased osmolarity, and relieved only when plasma osmolarity is decreased.
B) controlled by centers in the hypothalamus.
C) stimulated by decreased osmolarity.
D) relieved only when plasma osmolarity is increased.
E) controlled by centers in the hypothalamus and stimulated by increased osmolarity.

31) A hormone that helps to regulate the sodium ion concentration of the blood is 31) ________
A) cortisol.
B) somatotropin.
C) parathormone.
D) thymosin.
E) aldosterone.

32) Which about the hormone atrial natriuretic peptide is FALSE? 32) ________
A) increases sodium loss at the kidneys
B) decreases blood pressure
C) increases aldosterone secretion
D) produced by cells in the heart
E) decreases vasopressin secretion

33) The enzyme renin is responsible for the production of 33) ________
A) atrial natriuretic peptide.
B) erythropoietin.
C) angiotensinogen.
D) angiotensin I.
E) cortisol.

34) Angiotensin I is converted to angiotensin II by enzymes primarily located in the 34) ________
A) heart.
B) lungs and blood vessels.
C) kidneys.
D) liver.

35) The osmolarity in the bottom of the loop of Henle is ________ mOsM. 35) ________
A) 100
B) 900
C) 300
D) 1200
E) None of these answers are correct.

36) Decreased ECF volume causes 36) ________
A) the force of ventricular contraction to decrease.
B) sympathetic output from the cardiovascular control center to increase and arteriolar vasodilation.
C) sympathetic output from the cardiovascular control center to increase.
D) parasympathetic output from the cardiovascular control center to increase.
E) arteriolar vasodilation.

37) Symptoms of low plasma pH may include 37) _______
A) CNS depression only.
B) CNS depression and confusion and disorientation.
C) CNS depression; confusion and disorientation; and numbness, tingling, or muscle twitches.
D) confusion and disorientation only.
E) numbness, tingling, or muscle twitches only.

38) The most important factor affecting the pH of plasma is the concentration of 38) _______
A) hydrochloric acid.
B) carbon dioxide.
C) lactic acid.
D) ketone bodies.
E) organic acids.

39) The primary role of the carbonic acid–bicarbonate buffer system is to 39) _______
A) buffer carbonic acid formed by carbon dioxide.
B) buffer stomach acid.
C) buffer the urine.
D) increase the amount of carbonic acid during ventilation.
E) prevent pH changes caused by organic and fixed acids.

40) Which is most likely to be observed in a patient with compensated respiratory alkalosis? 40) _______
A) respiratory rate increases
B) kidneys reabsorb more hydrogen ions
C) kidneys conserve bicarbonate
D) body keeps less carbon dioxide
E) tidal volume increases

41) Vomiting of the stomach's contents can cause 41) _______
A) metabolic alkalosis.
B) respiratory alkalosis.
C) respiratory acidosis.
D) metabolic acidosis.
E) None of the answers are correct.

42) Emphysema can cause 42) _______
A) respiratory acidosis.
B) metabolic alkalosis.
C) metabolic acidosis.
D) respiratory alkalosis.
E) None of the answers are correct.

43) Hyperventilation can cause 43) _______
A) metabolic alkalosis.
B) respiratory alkalosis.
C) respiratory acidosis.
D) metabolic acidosis.
E) None of the answers are correct.

44) If sodium increases in the ECF, water will move from 44) _______
A) the ECF to cells, and cells will shrink.
B) cells to the ECF, and cells will shrink.
C) the ECF to cells, and cells will swell.
D) cells to the ECF, and cells will swell.

45) A buffer 45) _______
A) always decreases pH.
B) always increases pH.
C) binds or releases bicarbonate ions.
D) moderates changes in pH.

46) The normal pH range for most body fluids is 46) _______
A) 7.25 to 7.75.
B) 7 to 8.
C) 7.5 to 8.
D) 7.38 to 7.42.

47) When the pH rises above 7.42, a state of ________ exists. 47) _______
A) equilibrium
B) alkalosis
C) homeostasis
D) acidosis

48) Abnormal fat and amino acid metabolism may lead to the condition called 48) _______
A) ketoacidosis.
B) respiratory acidosis.
C) lactic acidosis.
D) metabolic alkalosis.

49) The enzyme that catalyzes the conversion of H_2O and CO_2 to H_2CO_3 is called 49) _______
A) carbonic anhydrase.
B) carbonic acid.
C) renin.
D) bicarbonate ion.

50) Normal removal of excess water in urine is known as 50) _______
A) diuretics.
B) osmotic diuresis.
C) filtration.
D) diuresis.

51) ________ interstitial osmolarity allows urine to be concentrated. 51) _______
A) Low medullary
B) High cortex
C) High medullary
D) Low cortex

52) AQP2 water pores are added to the cell membrane by ________ and withdrawn by ________ in a process known as ________. 52) _______
A) membrane recycling, endocytosis, exocytosis
B) membrane recycling, exocytosis, endocytosis
C) endocytosis, exocytosis, membrane recycling
D) exocytosis, endocytosis, membrane recycling

53) The anatomical arrangement of the kidney that allows transfer of solutes from one blood vessel to another is called the 53) _______
A) countercurrent exchange system.
B) portal system.
C) countercurrent heat exchanger.
D) capillaries.

54) Paracrine feedback from the ________ in the distal tubule to the granular cells stimulates release of ________. 54) _______
A) granular cells, renin
B) sympathetic neurons, epinephrine
C) liver, angiotensinogen
D) macula densa, renin

55) The ________ cells of the distal nephron are interspersed among the principal cells and contribute to acid-base regulation. 55) ________
A) endothelial B) intercalated C) granular D) endocrine

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

56) How do kidneys control urine concentration? 56) ______________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

57) Increased angiotensin II levels would result in increased 57) ________
A) blood volume.
B) retention of water.
C) retention of sodium ions at the kidney.
D) blood pressure.
E) All of these effects.

58) The RAS pathway begins with secretion of 58) ________
A) renin.
B) aldosterone.
C) angiotensin converting enzyme.
D) vasopressin.
E) angiotensinogen.

59) Renal tubule cells in the kidney medulla are constantly exposed to high extracellular osmolarity. How do they maintain normal cell volume? 59) ________
A) They synthesize organic solutes as needed to match the osmolarity.
B) They maintain a water-impermeable membrane.
C) They add or remove aquaporins as needed.
D) They synthesize water molecules through increased metabolism to offset volume loss.

60) When the pH of body fluids decreases, proteins will 60) ________
A) lose three-dimensional structure. B) fold into tertiary structures.
C) become more active. D) not be affected.

61) When the pH of the extracellular fluid decreases, the kidneys 61) ________
A) reabsorb more potassium ions.
B) excrete more sodium ions.
C) excrete more bicarbonate ions.
D) reabsorb more hydrogen ions.
E) reabsorb less water.

62) Two hours before major surgery, the patient is stressed, with increased heart rate and blood pressure. These symptoms are the result of 62) ________
A) increased parasympathetic activity.
B) sympathetic activation.
C) decreased activity of sympathetic centers in the hypothalamus.
D) decreased levels of epinephrine in the blood.
E) All of these mechanisms.

63) An explorer has been lost in the desert for two days with very little water. As a result, you would expect to observe 63) ________
A) cells enlarged with fluid.
B) increased vasopressin levels.
C) increased blood volume.
D) normal urine production.
E) decreased blood osmolarity.

64) Which effect would a decrease in pH have on the amount of potassium ion in the urine? 64) ________
A) increase
B) decrease
C) no effect

65) In response to a rapid increase of organic acid in the body, you would expect to observe 65) ________
A) decreased heart rate.
B) increased alveolar ventilation.
C) increased blood pH.
D) decreased blood pressure.

66) Dehydration may cause some ions to become concentrated. If a person was suffering from severe hyperkalemia, you would expect 66) ________
A) the membrane potential of nerves and muscles to be more negative.
B) the potassium ion concentration of the interstitial fluid to be less than normal.
C) muscle weakness and increased strength of twitch contractions.
D) abnormal cardiac rhythms.
E) All of the answers are correct.

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

67) Draw a map that shows the renal compensation for acidosis. Draw another map for alkalosis.

68) Diagram the reactions and interactions of the renin–angiotensin system (RAS). Which condition is the primary stimulus for its activation?

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

69) Which leads to the lowest water loss during a day? 69) ________
A) urine
B) feces
C) skin and lungs
D) metabolism
E) food and drink

70) The most potent stimulus for vasopressin release is 70) ________
A) low potassium.
B) blood volume.
C) plasma osmolarity.
D) blood pressure.

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

71) The cellular mechanisms for renal handling of H^+ and HCO_3^- involve several membrane transporters; name three.

72) What are the two mechanisms of bicarbonate reabsorption in the proximal tubule?

73) "Water, water, everywhere, nor any drop to drink" is a phrase from the Rime of the Ancient Mariner by Samuel Taylor Coleridge. This poem describes an ocean ship unable to sail to land and running out of its supply of fresh water; sailors have long known that drinking seawater cannot prevent death from dehydration. What would result if the sailors attempted to decrease their dehydration by drinking seawater? Defend your answer by describing kidney physiology. Why wouldn't reflexes, in response to dehydration, fully compensate? What does this show about the force allowing kidneys to retain water under more normal conditions?

74) What are oropharynx receptors, which hormone do they decrease, and how is it known that they exist? If a person stranded on a desert island drank seawater to try to quench his thirst, how would this affect the oropharynx receptors?

75) "Glucose, glucose, everywhere, nor any speck to utilize" is a phrase made up by one of the authors working on this test question. Similar to the irony of not being able to prevent dehydration by drinking seawater, people with untreated diabetes mellitus are unable to prevent starvation even though there is a large amount of glucose surrounding their cells; as if that is not bad enough, dehydration is also a problem. Explain why there is glucose in the urine of such people, why glucose is not present in the urine of normal people, and why diabetics become dehydrated.

76) Creating new food recipes is more successful when the chefs understand basic food chemistry. For example, to create a new cake recipe, one must understand the role of both acids and bases in leavening (rising, or trapping of tiny gas bubbles as dough is baking, to provide the characteristic texture). The principle is to add baking soda ($NaHCO_3$) into an acidic batter, causing a chemical reaction resulting in gas production. Looking at the chemical formula, which gas do you think is formed? Summarize the chemical reaction that occurs in a cake batter that contains $NaHCO_3$ and vinegar or acetic acid (produces acetate when ionized: $CH_3COO^- + H^+$). What would happen to your cake if you didn't use enough baking soda? What would happen if you forgot to acidify your batter with vinegar or a similar food acid?

77) Gopal suffers from chronic emphysema. Blood tests show that his pH is normal but his bicarbonate levels are increased significantly. How can this be?

78) Chandi, a nursing student, has been caring for burn patients. She notices that they consistently show increased levels of potassium in their urine and wonders why. What would you tell her?

79) Mr. Gregory comes to the doctor with high blood pressure. Tests show that he also has increased levels of renin in his blood and atherosclerotic plaques that have nearly blocked blood flow through his renal arteries. Mr. Gregory does not understand. Explain to him how decreased blood flow in his renal arteries could cause renin secretion to increase. Map the pathways through which increased renin causes high blood pressure for Mr. Gregory.

80) How does decreased blood pressure affect the following: granular cells, glomerulus, cardiovascular control center, hypothalamus? Indicate if decreased blood pressure directly affects the organ or tissue above or acts through a reflex pathway. If it acts through a reflex pathway, name the reflex.

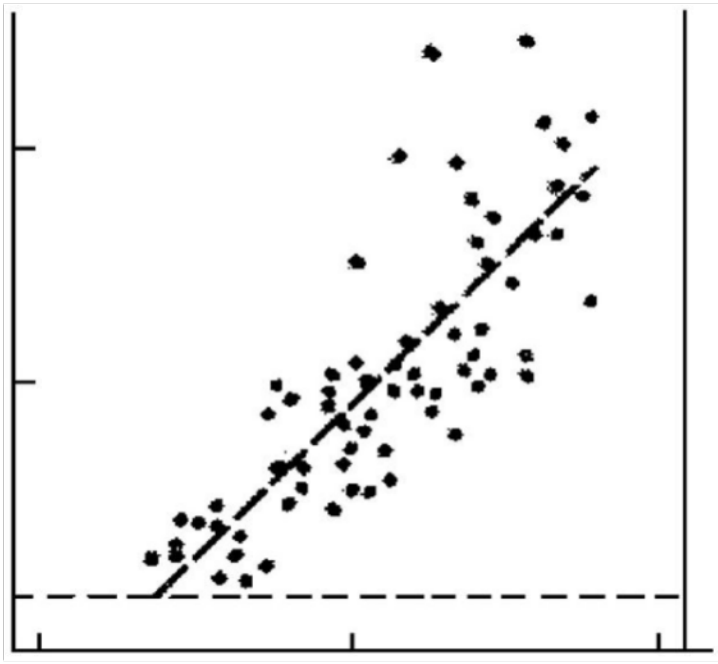
81) Layla, an undergraduate, has normal P_{CO_2} levels, high H^+ levels, low pH and bicarbonate levels. What type of disturbance is Layla suffering from and what might cause this? If her P_{CO_2} were elevated, would your answer change? Explain.

82) Diabetes mellitus produces many homeostatic imbalances, including acidosis. The pH imbalance is due to ketoacidosis, which results from excessive accumulation of by-products of fat metabolism, as the body cannot meet energy needs from carbohydrate metabolism. Maria is a teenaged diabetic who sometimes does not taking her insulin. Her mother takes her to the hospital because her breathing has become deep and gasping. Explain Maria's breathing pattern. Which other compensatory responses may occur, and would they occur earlier or later than the respiratory response?

83) Diabetes mellitus produces many homeostatic imbalances, including acidosis. The pH imbalance is due to ketoacidosis, which results from excessive accumulation of by-products of fat metabolism, as the body cannot meet energy needs from carbohydrate metabolism. Maria is a teenaged diabetic who sometimes does not taking her insulin. Her body is beginning to develop ketoacidosis as a result. Create a chart to indicate and explain how her blood pH, HCO_3^-, and P_{CO_2} react, by indicating "increase," "decrease," or "no change."

1. ketoacidosis has just developed
2. respiratory compensation occurs
3. renal compensation occurs

84) The graph below had its labels removed. One axis is plasma osmolarity. The other axis is EITHER plasma vasopı concentration OR plasma aldosterone concentration. Write the correct label on each axis of the graph, and explai: you came to that conclusion.



Answer Key
Testname: UNTITLED39

1) E
2) A
3) A
4) A
5) B
6) E
7) B
8) B
9) B
10) C
11) B
12) A
13) A
14) B
15) C
16) A
17) D
18) D
19) A
20) B
21) E
22) B
23) A
24) C
25) A
26) C
27) B
28) B
29) D
30) E
31) E
32) C
33) D
34) B
35) D
36) C
37) B
38) B
39) E
40) B
41) A
42) A
43) B
44) B
45) D
46) D
47) B
48) A
49) A
50) D

51) C
52) D
53) A
54) D
55) B
56) Kidneys control urine concentration by varying the amounts of water and sodium reabsorbed in the distal nephron.
57) E
58) A
59) A
60) A
61) A
62) B
63) B
64) B
65) B
66) D
67) Figure 20.18 shows compensations for both acidosis and alkalosis.
68) Figure 20.10 diagrams these interactions. The primary stimulus is low blood pressure, detected in several ways.
69) B
70) C
71) Possible answers include:
    1. apical Na^+–H^+ exchanger
    2. basolateral Na^+– HCO_3^- symport
    3. H^+– ATPase
    4. H^+– K^+– ATPase
    5. Na^+– NH_4^+ antiporter
72) 1. convert HCO_3^- into carbon dioxide, then back into HCO_3^-
    2. through the metabolism of glutamine (see Fig. 20.17)
73) The osmolarity of seawater is higher than that of the kidney medulla, thus the osmotic gradient that normally allows net reabsorption of water does not exist. Seawater osmolarity is largely a result of permeant ions such as sodium and chloride, so ingested seawater raises plasma osmolarity when it is absorbed. When the kidneys form a filtrate of this high–osmolarity solution, the descending limb fails to reabsorb water because the kidney medulla has a lower osmolarity than the filtrate (opposite of normal), along its entire length, and in fact the medulla would lose water as it moved into the descending limb. While ions would be reabsorbed by the ascending limb as usual, it is not enough to drive sufficient water reabsorption. Reflexes such as secretion of water–preserving hormones are ineffective because, ultimately, water reabsorption depends only on osmosis, and the normal osmotic gradient has been disrupted.
74) Oropharynx receptors inhibit the sense of thirst when they come in contact with cool water. They also inhibit secretion of vasopressin (ADH). They have not been identified anatomically, but presence of even a small amount of cold water in the mouth and pharynx is known to satisfy thirst even if the water is not absorbed and thus does not actually alleviate dehydration. Presence of receptors that inhibit the thirst sensation are the best guess as to how this works. Presumably drinking seawater would temporarily reduce the sense of thirst by the same proposed mechanism.
75) Glucose is in the urine simply because the plasma concentration exceeds the number of glucose transporters in the kidney tubules, because plasma glucose is unusually high. Normal individuals are able to reabsorb all the glucose in the filtrate because plasma glucose concentrations are normally low (normal body cells absorb glucose, diabetic cells do not). Glucose in the filtrate raises the osmolarity of the filtrate, which decreases the relative osmotic gradient for reabsorption of water; thus more water is lost in the urine leading to dehydration.

76) CO_2 gas is formed. $NaHCO_3 + CH_3COO^- + H^+ \rightarrow Na^+ + CH_3COO^- + CO_2 + H_2O$.
Less gas would be produced, so the cake wouldn't rise normally. The chemical reaction would not occur, so the cake wo rise.
77) As long as the ratio of bicarbonate ion to carbonic acid is 20:1, the pH of body fluids will remain normal. Since Gopal's condition is chronic (long term) his body has compensated for the excess carbonic acid (the result of hypercapnea due to poor ventilation) by increasing the amount of bicarbonate to match the elevated level of acid. This process involves the kidneys where some of the excess carbon dioxide is converted into carbonic acid and the carbonic acid is allowed to dissociate. The hydrogen ions are secreted and the newly formed bicarbonate is conserved to maintain a proper buffering capacity.
78) When tissues are burned, cells are destroyed and the contents of their cytoplasm leak into the interstitial fluid and then move into the plasma. Since potassium ion is normally found within the cell, damage to a large number of cells would release relatively large amounts of potassium into the blood. The elevated potassium levels would stimulate the cells of the adrenal cortex that produce aldosterone. The elevated levels of aldosterone would promote sodium retention and potassium secretion by the kidneys, thus accounting for the elevated levels of potassium in the patient's urine.
79) Atherosclerotic plaques block blood flow, which decreases GFR and decreases pressure in the afferent arteriole. These are both stimuli for renin release. Renin secretion starts a cascade that produces angiotensin II, a vasoconstrictor. Vasoconstriction increases blood pressure, and the medullary control center responds to ANG II by also increasing ADH, aldosterone secretion, and thirst, collectively increasing blood pressure even further.
80) See Table 20.1 in the chapter.
81) Layla is likely suffering from metabolic acidosis, since his P_{CO_2} is normal. If her P_{CO_2} was elevated, then she would likely be suffering from respiratory acidosis. Metabolic acidosis may be caused by ingestion of methanol, aspirin, and ethylene glycol. Respiratory acidosis can be caused by drugs or alcohol depression, increased resistance in asthma, impaired gas exchange in fibrosis or severe pneumonia, and muscle weakness in muscular dystrophy and other muscle diseases. The most common cause is chronic obstructive pulmonary disease.
82) Maria's respiratory system is responding to her ketoacidosis. By hyperventilating, she removes more CO_2 from her blood, which shifts the carbonic acid equation to the right, removing free H^+ and raising pH. Chemical buffering in the blood occurs immediately after the initial pH disturbance, as this is an ongoing process. The kidneys buffer more slowly, so they are effective later.
83) 1. pH decreases by definition, HCO_3^- decreases as the carbonic acid reaction shifts to the right, P_{CO_2} is unchanged because the excess is expelled by the lungs due to the higher gradient.
2. pH increases and HCO_3^- decreases as exhaled CO_2 drives the carbonic acid reaction to the right, and P_{CO_2} decrease as it is exhaled.
3. pH increases as H^+ is excreted, HCO_3^- increases as it is reabsorbed, and P_{CO_2} increases as respiration returns to normal.
84) The dependent variable is typically used for the y-axis, and the independent variable for the x-axis. Plasma osmolarity is the independent variable, and it regulates the secretion of hormones. As osmolarity increases, vasopressin secretion increases to promote water retention, but aldosterone secretion is inhibited to promote sodium loss. Therefore vasopressin concentration is the correct label for the y-axis. See Figure 20.6 in the chapter.